



# **Relation between Geometry and Line Fluxes using CLASS data by XRF Techniques**

by

**Sanjay R**  
(2024PGPHPH10)  
NIT Jamshedpur

A thesis submitted in partial fulfilment for the  
degree of Master of Science to the

**Department of Physics**  
National Institute of Technology  
Jamshedpur, Jharkhand  
May 2026

Under the esteemed Guidance of

**Dr Shyama Narendranath K C**  
Sci / Eng – SG  
Space Astronomy Group (SAG)  
URSC Bengaluru

**Dr Abhishek Majhi**  
Assistant Professor  
Department of Physics  
NIT Jamshedpur

**Mrs Netra S Pillai**  
Sci / Eng – SD  
Space Astronomy Group (SAG)  
URSC Bengaluru



Department of Physics  
National Institute of Technology, Jamshedpur  
An Institute of National Importance under Ministry of Education, Govt. of India Certificate

## Certificate

This is to certify that the thesis entitled ***“Relation between Geometry and Line Fluxes using CLASS data by XRF Techniques”***, which is being submitted by **Mr. Sanjay R** with Registration No. **2024PGPHPH010**, to the Department of Physics, National Institute of Technology, Jamshedpur, for the award of degree of Master of Science in Physics, is a record of research work carried out by him under our guidance. He has fulfilled the requirements for the submission of thesis, which to our knowledge has reached the requisite standard. The results contained in this dissertation have not been submitted in part or in full to any university or institute for the award of any degree.

Dr Shyama Narendranath K C  
Sci /Eng - SG  
Space Astronomy Group (SAG)  
URSC Bengaluru

Dr Abhishek Majhi  
Assistant Professor  
Department of Physics  
NIT Jamshedpur

Netra S Pillai  
Sci/Eng – SD  
Space Astronomy Group (SAG)  
URSC Bengaluru.

Date:  
Place:

# Declaration of Authorship

I hereby declare that the MSc thesis titled “*Relation between Geometry and Line Fluxes using CLASS data by XRF Techniques*” is an authentic record of the research work carried out by me under the supervision of Dr. Shyama Narendranath, SAG, URSC Bengaluru, Dr Abhishek Majhi, Department of Physics, National Institute of Technology Jamshedpur, Jharkhand and Mrs Netra S Pillai SAG, URSC, Bengaluru No part of this work has formed the basis for award of any other degree or diploma in any university or institution.

Sanjay R  
2024PGPHPH110

May 2026

# Abstract

This thesis presents an empirical investigation into the relationship between X-ray fluorescence (XRF) line fluxes and geometry mainly incidence angle utilising observations from the CLASS (Chandrayaan-2 Large Area Soft X-ray Spectrometer) instrument aboard ISRO's Chandrayaan-2 orbiter. The work integrates two complementary methodologies: laboratory-based XRF measurements on reference samples and orbital spectral data analysis of CLASS observations acquired during solar flare enhancement periods in April 2022. In the laboratory phase, a controlled XRF experimental setup was employed to measure the angular dependence of characteristic X-ray line intensities from known standard samples. Line fluxes systematically determined as a function of measurement geometry, establishing a ground-truth angular-flux relationship. These measurements directly informed the geometric correction framework applied to the orbital data. In the space-based analysis, raw CLASS spectrogram data (14 FITS files; 100-second temporal cadence) were stacked, temporally cross-matched with solar flare enhancement catalogues, and subjected to Gaussian spectral deconvolution within calibrated energy windows ( $13.5 \text{ eV channel}^{-1}$ ). Fundamental spectral parameters — line centroid, amplitude, and width were extracted for the  $K\alpha$  emission lines of Mg, Al and Si. Integrated line areas (cps) served as the primary line flux proxy, with event quality classification (good/bad) applied on the basis of centroid accuracy, and goodness-of-fit criteria. Solar zenith angle corrections were applied using co-registered SPICE ephemeris data to normalise illumination geometry across orbital tracks. The corrected line flux derived from CLASS good events were plotted against solar Incidence angle and cross-correlated with known elemental abundances. As well as with the data provided find the highland and mare region present in the moon. The combined laboratory-space methodology validates the angular dependence of XRF line flux in both experimental and orbital regimes and demonstrates that CLASS flare-time measurements carry quantitatively meaningful compositional information. The empirical calibration relations established in this work provide a foundation for quantitative analysis between the line fluxes and geometry of the lunar regolith using CLASS orbital data.

# Acknowledgement

A project successfully completed does not showcase the untiring effort of the project trainee alone. It is also the manifestation of the invaluable guidance, appropriate suggestion and encouragement extended by my guides, professors and well-wishers. Their whole hearted support and constant guidance has crowned the effort worth success.

**“Relation between Geometry and Line Fluxes using CLASS data by XRF Techniques”** has been a pleasant experience for me. I met outstanding scientists at ISRO whom I had a lot to learn from. They have encouraged and helped us during the project. It has been an excellent experience to work in such a big Organisation

I would like to thank my HOD Dr Hira Lal Yadav for extending his helping hand at times of need. I thank Dr Abhishek Majhi my internal guide for his timely suggestion and sound advice

I am extremely thankful to HRDD(URSC) for giving me this opportunity to carry my project. I am very thankful for having Dr Shyama Narendranath scientist (ISRO) and Mrs Netra S Pillai (ISRO) as my external guide in the organisation. In spite of their busy schedule they took time off and dedicatedly helped me in completing my project within the time schedule.

Words are not sufficient to thank my Amma, Appa and my cute Thangachima who have been my constant source of inspiration and encouragement in all my endeavours

# Content

## Chapter I

|                                  |          |
|----------------------------------|----------|
| <b>1 Introduction.....</b>       | <b>9</b> |
| 1.1 Background.....              | 9        |
| 1.2 Purpose and Objective .....  | 10       |
| 1.3 Scientific significance..... | 11       |
| 1.4 Key words .....              | 12       |
| 1.5 Scope and Limitation .....   | 12       |
| 1.6 Thesis Outline .....         | 12       |

## Chapter II

|                                                                         |           |
|-------------------------------------------------------------------------|-----------|
| <b>2 Literature Review .....</b>                                        | <b>13</b> |
| 2.1 X-Rays .....                                                        | 13        |
| 2.2 Production of X-rays.....                                           | 13        |
| 2.3 Interaction of x-rays with matter.....                              | 14        |
| 2.4 Binding energy of shells .....                                      | 16        |
| 2.5 X-RAY SCATTERING .....                                              | 16        |
| 2.5.1 Coherent Scattering.....                                          | 17        |
| 2.5.2 Incoherent Scattering .....                                       | 17        |
| 2.5.3 Comparison between coherent and incoherent scattering.....        | 18        |
| 2.6 Production of characteristic X-ray fluorescent radiation (XRF)..... | 19        |
| 2.7 XRF Classification.....                                             | 20        |
| 2.8 Mechanism of XRF .....                                              | 21        |
| 2.9 Derivation of primary XRF intensity .....                           | 22        |
| 2.10 Factors affecting XRF .....                                        | 23        |

## Chapter III

|                                                                   |    |
|-------------------------------------------------------------------|----|
| 3.1 Chandrayaan II.....                                           | 26 |
| 3.2 Chandrayaan-2 Large Area Soft X-ray Spectrometer (CLASS)..... | 27 |
| 3.2.1 Instrument Configuration.....                               | 27 |

|                                |    |
|--------------------------------|----|
| 3.2.2 Working Principle.....   | 28 |
| 3.2.3 Detector System.....     | 28 |
| 3.2.4 Thermal Management.....  | 28 |
| 3.2.6 Data analysis.....       | 29 |
| 3.2.7 CLASS Data products..... | 29 |
| 3.3 Role of XSM.....           | 29 |

## **Chapter IV**

|                                                                 |    |
|-----------------------------------------------------------------|----|
| 4.1 Experimental setup.....                                     | 31 |
| 4.2 Experimental Procedure & Working Principle.....             | 33 |
| 4.3 Beam Geometry.....                                          | 34 |
| 4.4 Samples Used.....                                           | 35 |
| 4.5 Experimental Conditions and Precautions.....                | 35 |
| 4.6 Data Acquisition Procedure.....                             | 36 |
| 4.6.1 Measurement Protocol.....                                 | 36 |
| 4.6.2 MCA and Channel Number Assignment.....                    | 36 |
| 4.7 Peak Identification and Gaussian Fitting.....               | 37 |
| 4.7.1 Initial Spectral Inspection.....                          | 37 |
| 4.7.2 Physical origin of Fe K $\alpha$ and K $\beta$ Lines..... | 37 |
| 4.7.3 Gaussian Fitting.....                                     | 38 |
| 4.8 Energy Calibration.....                                     | 40 |
| 4.8.1 Channel to Energy Conversion.....                         | 40 |
| 4.8.2 Determining Gain and Offset.....                          | 40 |
| 4.9 Flux Determination.....                                     | 41 |
| 4.9.1 Line Flux.....                                            | 41 |
| 4.9.2 Poission Statistics for error analysis.....               | 42 |
| 4.10 Applying to all Samples.....                               | 42 |
| 4.11 Angular dependence of line flux.....                       | 46 |

## **Chapter V**

|                      |    |
|----------------------|----|
| 5.1 Solar Flare..... | 53 |
|----------------------|----|

|                                                          |    |
|----------------------------------------------------------|----|
| 5.2 Methodology .....                                    | 54 |
| 5.2.1 Spectral Stacking .....                            | 54 |
| 5.2.2 Flare Event Identification .....                   | 54 |
| 5.3 Energy Calibration .....                             | 55 |
| 5.4 Good vs Bad Event Classification .....               | 57 |
| 5.5 Adaptive Window Method.....                          | 58 |
| 5.6 Geometry merging .....                               | 58 |
| 5.7 Analysis.....                                        | 59 |
| 5.8 Data Integration and Preprocessing (in wt%) .....    | 60 |
| 5.8.1 Reference Composition.....                         | 61 |
| 5.8.2 Distance in Multi element Compositional Space..... | 61 |
| 5.8.3 Threshold on Measurement uncertainty.....          | 61 |
| 5.8.4 Classification Criterion .....                     | 62 |
| 5.8.5 Shoelace Formula.....                              | 62 |
| 5.9 Incidence Angle vs Line Flux .....                   | 63 |
| 5.10 Spatial Distribution of Mg & Al abundance .....     | 65 |
| 6 Result and Conclusion .....                            | 67 |
| 7. Future Scope and Research Direction.....              | 68 |
| 7.1 Solar Flux Normalisation Using XSM.....              | 68 |
| 7.2 Extension to Higher Solar Activity Periods.....      | 68 |
| 7.3 Full elemental Abundance Map Generation .....        | 69 |
| 7.4 Future Planetary Missions.....                       | 69 |
| 8. Reference .....                                       | 70 |

# Chapter - I

## 1.Introduction

The **MOON** - Earth's only natural satellite is considered to have originated from a giant impact with a Mars-sized body during the early years of formation of the solar system approximately 4.5 billion years ago. It is a body devoid of any significant atmosphere and hence preserves the rich history of the formation and evolution process of the solar system through its surface features and composition. Composition of the lunar surface involves identification of elements and minerals present. While minerals reveal the formation history and evolution of the crust, direct detection of elements that are the building blocks of minerals, is critical to decipher degeneracy in mineral signatures and constraining the formation models of the bulk Moon as well as of the various types of lithologies. Also interesting is the inevitable linkage of surface elemental abundances to atoms in the tenuous exosphere. The Moon's surface composition has been determined broadly as consisting of regions of enhanced levels of plagioclase (typical bright highlands) and basalts (typical dark mare regions). A significant portion of the lunar surface is covered by regolith, a layer of fine, fragmented material generated through continuous meteoritic impacts. This regolith plays a vital role in X-ray fluorescence (XRF) studies, as it interacts directly with incident solar X-rays. Due to the absence of an atmosphere, solar X-rays penetrate the surface without attenuation and excite atoms within the regolith, leading to the emission of characteristic X-ray lines. These emitted X-rays provide direct information about the elemental composition of the lunar surface, forming the basis for remote geochemical analysis. Instruments such as the Chandrayaan-2 Large Area Soft X-ray Spectrometer (CLASS) onboard the Chandrayaan-2 mission utilize this principle to measure the relation.

### 1.1 Background

The study of elemental composition in geological and extraterrestrial materials has been one of the central pursuits of analytical physics for over a century. Among the many techniques developed for this purpose, X-ray Fluorescence (XRF) spectroscopy stands out as one of the

most powerful, non-destructive, and versatile analytical methods available. By exploiting the quantum mechanical interaction between incident X-ray photons and the inner-shell electrons of target atoms, XRF enables the simultaneous identification and quantification of multiple elements across a wide range of concentrations — from major constituents down to trace levels in the parts-per-million (ppm) regime.

Over the past several decades, XRF has evolved from a laboratory-based bulk analysis technique into an instrument deployed aboard planetary exploration spacecraft. Missions such as NEAR Shoemaker (asteroid Eros, 1999-2001), ESA's SMART-1 lunar orbiter (2004-2006), JAXA's Kaguya/SELENE mission (2007-2009), NASA's MESSENGER at Mercury (2004-2015), and India's Chandrayaan-1 (2008-2009) and Chandrayaan-2 (2019-present) have all carried X-ray spectrometers to map elemental abundances on planetary surfaces. In the space context, solar X-rays illuminate the surface, inducing fluorescence from the top few micrometres of the regolith, and the emitted spectrum is recorded by detectors aboard the orbiting spacecraft.

The present thesis project is concerned with a fundamental problem at the heart of planetary XRF spectroscopy: establishing a robust, relation between geometry and Line fluxes. This problem is nontrivial because the detected flux from any given spectral line is modulated by a complex interplay of factors — the incident solar irradiance spectrum, the geometry of illumination and detection, matrix effects arising from the composition of the sample itself, and instrumental response functions. The CLASS (Chandrayaan-2 Large Area Soft X-ray Spectrometer) instrument aboard ISRO's Chandrayaan-2 orbiter provides the real-world data context for this investigation.

## **1.2 Purposes and Objectives**

The primary purpose of this research is to develop and validate an empirical framework that translates observable X-ray spectral quantities — specifically, the ratios of characteristic line fluxes — into quantitative elemental abundance estimates expressed in weight percent (wt%). Such a framework is essential for the scientific interpretation of CLASS data and for extending XRF-based geochemistry to future planetary missions.

The specific objectives of this thesis are as follows:

1. To design and implement a laboratory XRF experimental setup capable of measuring characteristic X-ray line fluxes from elemental and compound targets across a

controlled range of angular geometries (10° to 70° in 5° steps), using a Silicon Drift Detector for energy-dispersive detection.

2. To collect angular-resolved X-ray fluorescence spectra from samples including mild steel, Mu metal, Steel, Copper, Zinc Selenide (ZnSe), Ti -6-4 and JSC-1A lunar regolith to extract characteristic line fluxes through Gaussian spectral fitting and energy calibration.
3. To process real CLASS observational data from the Chandrayaan-2 mission: handling raw FITS files, performing event-by-event analysis to discriminate good from bad events based on grade selection and quality flags, stacking spectra for improved signal-to-noise, and deriving calibrated line fluxes for lunar surface regions.
4. To correlate the measured or computed XRF line flux ratios with independently known elemental abundances (wt%) from standard geological reference materials and CLASS normalisation data, establishing regression-based empirical relationships.
5. To validate the empirical relationships by comparing predicted and actual compositions and to assess the sensitivity, accuracy, and limitations of the method in the context of both laboratory and orbital measurements.

### **1.3 Scientific Significance**

Quantitative elemental mapping of planetary surfaces is fundamental to understanding planetary formation and evolution. The Moon, as Earth's nearest neighbour and the only extraterrestrial body visited by humans, preserves a record of the early solar system. Its surface composition reflects processes of magmatic differentiation, impact gardening, and space weathering operating over 4.5 billion years. Accurate XRF-based abundance maps from CLASS can constrain models of the lunar magma ocean, identify compositional diversity between the ancient feldspathic highlands and the younger volcanic mare basalts, and contribute to resource assessment for future lunar exploration and in-situ resource utilisation (ISRU).

Beyond the Moon, the methodology developed here — empirical line-flux-ratio to abundance calibration — is directly transferable to future XRF experiments planned for Mars, Mercury, asteroids, and other airless bodies

### **1.4 Key Words**

X-ray Fluorescence (XRF), CLASS - (Chandrayaan-2 Large Area Soft X-ray Spectrometer), FITS (Flexible Image Transport System), Silicon Drift Detector (SDD)

## 1.5 Scope

The scope of this work encompasses the laboratory calibration phase(angular phase), the observational data processing phase(polar phase), and the empirical calibration derivation phase - though the methodology developed here provides a direct pathway towards such maps. The angular range studied in the laboratory ( $10^{\circ}$ – $70^{\circ}$  in  $5^{\circ}$  increments) was selected to bracket the typical solar zenith angle (Incident beam angle -  $90^{\circ}$ ) and emission angles encountered in orbital XRF geometry.

## 1.6 Thesis Outline

The remainder of this thesis is structured as follows:

- Chapter 2 – Literature Review: Surveys the theoretical foundations of XRF physics, the historical development of planetary XRF spectroscopy, and the specific scientific context of the Chandrayaan missions. Prior work on empirical XRF calibration methodologies
- Chapter 3 – **Chandrayaan-2 Mission** Provides an overview of the Chandrayaan-2 mission and its associated satellite systems, focusing on the CLASS instrument and the supporting XSM payload, and their significance in X-ray fluorescence studies..
- Chapter 4 – Laboratory Experimental Setup and Data Acquisition: Describes the design and alignment of the laboratory XRF setup, Spectral Analysis and Energy Calibration: Presents the methodology for Gaussian fitting of spectral lines, energy calibration procedures, background subtraction, and the extraction of calibrated line fluxes and their uncertainties.
- Chapter 5– CLASS Data Processing: Details the pipeline for handling Chandrayaan-2 CLASS FITS files, including event-level quality selection (good/bad event discrimination), grade filtering, spectrum stacking, and the derivation of orbital line fluxes.
- 6 Results: Presents the central results of the
- 7 Future Scope and Design: Summarises the key findings and identifies directions for future research.
- 8 References

# Chapter II

## 2.1 X-Rays

X-rays are discovered in 1895 by the German physicist Wilhelm Conrad Rontgen while experimenting with cathode rays. X-rays are a form of high-energy electromagnetic radiation with wavelengths typically ranging from  $10^{-12}$  to  $10^{-8}$  meters, corresponding to energies between approximately 0.1 keV and 125 keV[4]. They occupy the region of the electromagnetic spectrum between ultraviolet radiation and gamma rays.

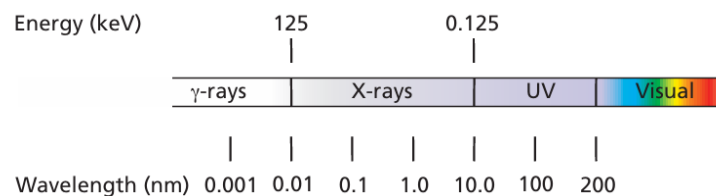


Figure 2.1 X-rays and other electromagnetic radiation[5]

The wavelength of an X-ray is inversely proportional to its energy and is related through the equation

$$E\lambda = hc$$

where  $E$  represents the photon energy in keV,  $\lambda$  represents the wavelength in nm,  $h$  is Planck's constant, and  $c$  is the speed of light. The product  $hc$ , when expressed in units of keV·nm, has a constant value of 1.23985. Due to their high energy and penetrating power, X-rays are capable of interacting with matter at the atomic level, making them highly suitable for probing the internal structure and composition of materials.

## 2.2 Production of X-rays

When a high voltage is applied across the electrodes, electrons are thermionically emitted from the heated cathode and accelerated towards the anode (target material), usually made of high atomic number elements such as tungsten. Upon striking the target, the kinetic energy of the electrons is converted into X-ray radiation through two primary mechanisms:

Bremsstrahlung radiation

Characteristic X-ray emission [5].

Bremsstrahlung radiation

meaning “braking radiation,” arises when incident electrons are decelerated in the Coulomb field of atomic nuclei within the target material. This deceleration results in the emission of a continuous spectrum of X-ray energies, extending from zero up to a maximum value determined by the energy of the incident electrons. The maximum photon energy corresponds to the applied accelerating voltage, such that

$$E_{\max} = eV,$$

where  $V$  is the tube voltage. This continuous spectrum forms the background in X-ray spectra and is strongly dependent on both the accelerating voltage and the atomic number of the target [7].

#### Characteristic X-ray emission

occurs when high-energy electrons eject inner-shell electrons (typically from the K or L shells) of the target atoms. The resulting vacancy is subsequently filled by an electron from a higher energy level, leading to the emission of X-rays with discrete energies that are characteristic of the specific element. These emissions are classified as  $K\alpha$ ,  $K\beta$ ,  $L\alpha$ , etc.,

$$E = E_{\text{higher}} - E_{\text{lower}}$$

depending on the electronic transition involved. Unlike Bremsstrahlung radiation, characteristic X-rays produce sharp spectral lines that are uniquely associated with individual elements and form the basis for elemental identification in X-ray fluorescence (XRF) analysis.

## 2.3 Interaction of x-rays with matter

When X-rays interact with matter, two processes occur: Scattering and the Photoelectric effect. Scattering occurs when X-rays deviate from their original path due to interactions with atomic electrons, while the photoelectric effect involves the complete absorption of an X-ray photon followed by the ejection of a bound electron (photo-electron).

Scattering can be either coherent (elastic, no energy loss) or incoherent (Compton scattering, energy loss and wavelength shift). The photoelectric effect dominates at lower X-ray energies, especially near atomic absorption edges, and is the basis for X-ray fluorescence (XRF).

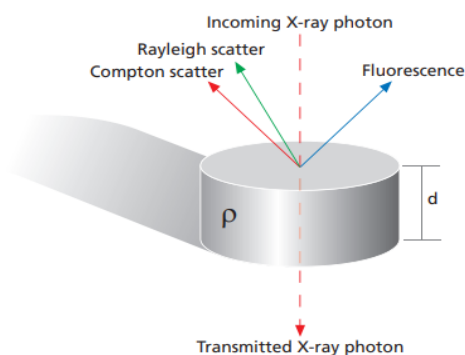


Fig : Interaction of X rays with matter [3]

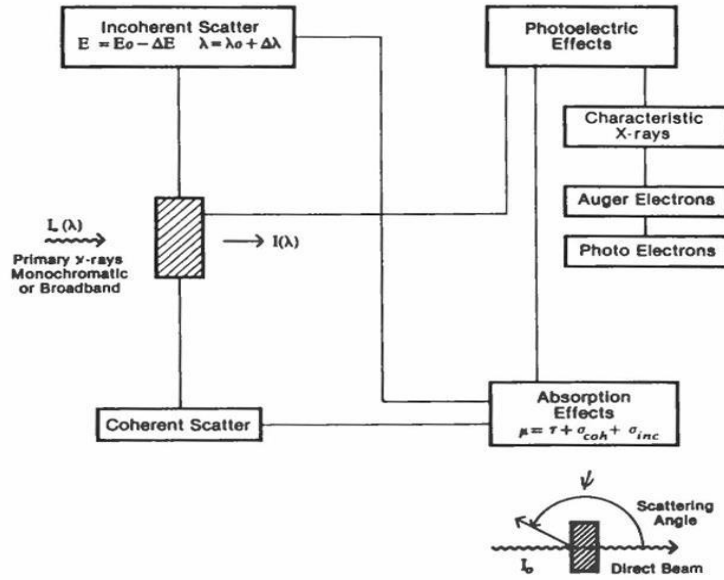


Fig 1.2 Interaction of X-rays with matter (source: Quantitative X-ray Spectrometry)[6]

The Beer- Lambert law describes the exponential attenuation of X-ray intensity ( $I$ ) through a material:

$$I = I_0 e^{-\mu x}$$

where:

- $I_0$  = initial intensity,
- $\mu = \tau + \sigma_{coh} + \sigma_{incoh}$  (total mass attenuation coefficient, combining photoelectric ( $\tau$ ), coherent ( $\sigma_{coh}$ ), and incoherent ( $\sigma_{incoh}$ ) scattering contributions).

The mass attenuation coefficient quantifies how X-rays are absorbed or scattered as they pass through a material. For mixtures, the total  $\mu$  is the weighted sum of each element's attenuation coefficient:

$$\mu_{total} = \sum W_j \mu_j$$

where  $W_j$  is the weight fraction of element  $j$ . Discontinuities in  $\mu$  (absorption edges) occur at energies matching electron binding energies (e.g., K-edge), critical for selective excitation in XRF.[7]

The factors affecting mass attenuation coefficient are:

- Energy dependence:  $\mu$  decreases with increasing photon energy, except at absorption edges.
- Atomic structure: Tightly bound inner-shell electrons (K, L) dominate photoelectric absorption, while outer electrons contribute to scattering.
- Specimen composition: Light elements (e.g., carbon) exhibit higher scattering/background, while heavy elements (e.g., lead) absorb strongly.[8]

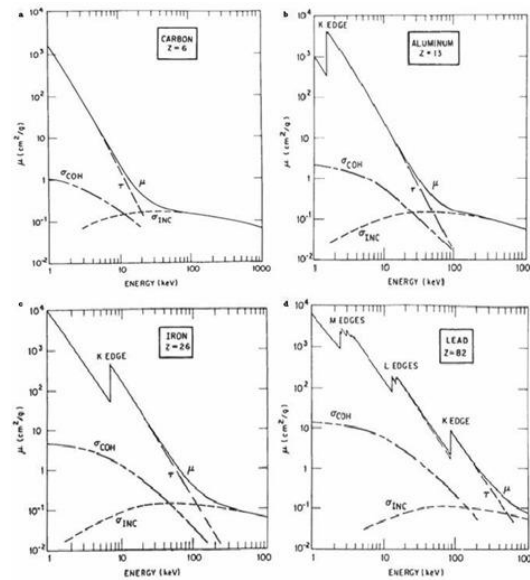


Fig 1.3 Mass attenuation coefficient and its components as a function of incident photon energy for carbon(a), aluminium(b), iron(c) and lead(d) (source: Quantitative X-ray Spectrometry)[6]

## Binding energy of shells

Atoms have a nucleus surrounded by electrons arranged in shells (K, L, M, etc.), each with specific energy levels and capacities defined by  $2n^2$ . These shells contain subshells (s, p, d, f), further divided into orbitals. Electron arrangement follows four quantum numbers: principal (n), azimuthal (l), magnetic (m), and spin (s), with electrons filling orbitals in order of increasing energy according to the Aufbau principle. Each orbital holds up to two electrons with opposite spins. This electron configuration determines an element's behaviour. In X-ray spectroscopy, absorption edge refers to the minimum energy needed to eject electrons from specific shells, increasing with atomic number (Z) as described by Moseley's law:

$$E \propto (Z - \sigma)^2$$

| Shell | n (Principal Quantum Number) | l (Azimuthal Quantum Number) | m (Magnetic Quantum Number) |
|-------|------------------------------|------------------------------|-----------------------------|
| K     | 1                            | 0                            | 0                           |
| L1    | 2                            | 0                            | 0                           |
| L2    | 2                            | 1                            | -1, 0, +1                   |
| L3    | 2                            | 1                            | -1, 0, +1                   |
| M1    | 3                            | 0                            | 0                           |
| M2    | 3                            | 1                            | -1, 0, +1                   |
| M3    | 3                            | 1                            | -1, 0, +1                   |
| M4    | 3                            | 2                            | -2, -1, 0, +1, +2           |
| M5    | 3                            | 2                            | -2, -1, 0, +1, +2           |

## 2.4 X-RAY SCATTERING

X-ray scattering refers to the phenomenon where X-ray photons interact with matter and are deflected from their original path. This interaction can occur in two primary ways:

coherent (Rayleigh) scattering and incoherent (Compton) scattering. The factors affecting scattering include the energy of the incident X-rays, the atomic number of the scattering material, the scattering angle, and the electron density of the material. [9] Coherent scattering involves no energy loss and is dominant at lower energies, while incoherent scattering results in energy loss and becomes significant at higher energies.[10]

### 2.4.1 Coherent Scattering

Coherent scattering, also known as Rayleigh scattering, occurs when X-ray photons interact with bound electrons in an atom, causing the electrons to oscillate and re-emit

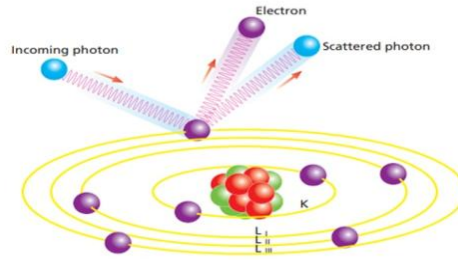


Fig Coherent Scattering [3]

photons of the same energy but in different directions. The differential cross-section (DCS) for coherent scattering from an atom is derived from Thomson scattering, which describes the scattering of X-rays by a free electron.[11]

$$\frac{d\mu_{(coh/inc)i}}{d\Omega} = \frac{N}{A_i} \frac{d\sigma_{(coh/inc)i}}{d\Omega},$$

where  $N$  is Avogadro constant ( $6.022 \times 10^{23} \text{ mol}^{-1}$ ),  $A_i$  is atomic mass of the  $i^{\text{th}}$  element,  $\sigma_i$  is cross section of scattering for a single atom of the  $i^{\text{th}}$  element.

In coherent scattering, the energy of the incident photon is equal to the energy of coherently scattered photon (i.e.  $E = E'$ ). Differential cross-section for an unpolarized photon is:

$$\frac{d\sigma_{(coh)i}}{d\Omega} = \frac{r_e^2}{2} (1 + \cos^2 \alpha) f_i^2$$

where  $r_e$  is classical electron radius =  $2.82 \times 10^{-13} \text{ cm}$ .  $\alpha$ : Scattering angle (angle between incident and scattered directions).  $f_i$ : Atomic scattering factor for the  $i^{\text{th}}$  element, which accounts for interference effects due to electron distribution. [12]

### 2.4.2 Incoherent Scattering

Incoherent scattering, or Compton scattering, occurs when X-ray photons interact with loosely bound or free electrons, resulting in a loss of energy and a change in wavelength. It is caused by the particle nature of photons and electrons. Flux of incoherent scattering is small relative to coherent scattering.

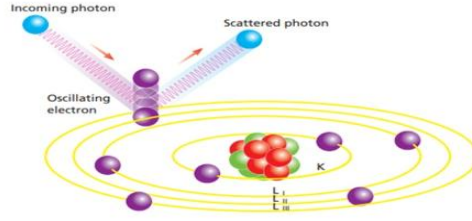


Fig Incoherent Scattering[3]

The differential cross-section for incoherent scattering is given by:[13]

$$\frac{d\sigma_{(inc)i}}{d\Omega} = \frac{d\sigma_{(KN)}}{d\Omega} S_i(E, \alpha)$$

$$\frac{d\sigma_{(KN)}}{d\Omega} = \frac{r_e^2}{2} \left(\frac{E_c}{E}\right)^2 \left(\frac{E_c}{E} + \frac{E}{E_c} - \sin^2 \alpha\right)$$

where  $d\sigma_{(KN)}/d\Omega$  is the Klein-Nishina cross-section for a free electron and  $S_i(E, \alpha)$  is the incoherent scattering function. The energy of the scattered photon is reduced to the Compton energy  $E_c$ :[14]

$$E_c = E' = \frac{m_e c^2 E}{m_e c^2 + E(1 - \cos \alpha)}$$

where  $m_e$  is the electron mass and  $c$  is the speed of light.

The corresponding change in wavelength is given by the Compton shift:

$$\Delta\lambda = \lambda' - \lambda = \frac{h}{m_e c} (1 - \cos \alpha)$$

where,  $h$  is Planck's constant ( $6.626 \times 10^{-34}$  kg m<sup>2</sup>/s).

### 2.4.3 Comparison between coherent and incoherent scattering

Coherent Scattering occurs when X-ray photons interact with tightly bound electrons in an atom, causing the electrons to oscillate and re-emit radiation at the same energy and wavelength as the incident photon. This elastic process is dominant at lower X-ray energies and is highly dependent on the atomic number ( $Z$ ) of the material. The angular distribution of coherently scattered X-rays follows a  $(1 + \cos 2\theta)$  pattern, peaking in the forward direction. In X-ray fluorescence (XRF) analysis, coherent scattering contributes to the background signal but does not distort peak positions, making it less problematic for qualitative analysis.[15] Incoherent Scattering, on the other hand, involves energy transfer to loosely bound or free electrons, resulting in a wavelength shift (Compton effect) and a broad continuum background in spectra. The energy loss is described by the Klein-Nishina formula, and the shift in wavelength ( $\Delta\lambda$ ) depends on the scattering angle ( $\theta$ ):[16]

$$\Delta\lambda = \lambda' - \lambda = \frac{h}{m_e c} (1 - \cos \alpha)$$

Unlike coherent scattering, incoherent scattering scales linearly with  $Z$  and becomes significant at higher X-ray energies. Its angular distribution is more isotropic, though it shifts toward larger angles as energy increases.[17]

## 2.5 Production of characteristic X-ray fluorescent radiation (XRF)

We know the classical model of an atom is a nucleus with positively charged protons and non-charged neutrons, surrounded by electrons grouped in shells or orbitals. The innermost shell is called the K-shell, followed by L-shells, M-shells etc. as one moves outwards. The L-shell has 3 sub-shells called  $L_I$ ,  $L_{II}$  and  $L_{III}$ . The K-shell can contain 2 electrons, and the L-shell contains 8 electron. The energy of an electron depends on the shell it occupies, and on the element to which it belongs. When irradiating an atom, particles such as X-ray photons with sufficient energy can expel an electron from the atom. This produces a 'hole' in a shell, putting the atom in an unstable excited state with a higher energy. The 'hole' in the shell is also called the initial vacancy[3]. The atom wants to restore the original configuration, and this is done by transferring an electron from an outer shell such as the L-shell to the hole in the K-shell. An L-shell electron has a higher energy than a K-shell electron, and when an L-shell electron is transferred to the K-shell, the energy surplus can be emitted as an X-ray photon. In a spectrum, this is seen as a line. The energy of the emitted X-rays depends on the difference in energy between the shell with the initial hole and the energy of the electron that fills the hole (in the example, the difference between the energy of the K and the L shell). Each atom has its specific energy levels, so the emitted radiation is characteristic of that atom. An atom emits more than a single energy (or line) because different holes can be produced and different electrons can fill these collection of emitted lines is characteristic of the element and can be considered a fingerprint of the element.

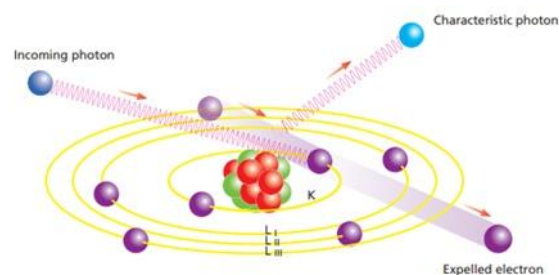


Fig. Production of characteristic radiation[3]

To expel an electron from an atom, the X-rays must have a higher energy level than the binding energy of the electron. If an electron is expelled, the incoming radiation is absorbed, and the higher the absorption the higher the fluorescence. If, on the other hand, the energy is too high, many photons will 'pass' the atom and only a few electrons will be removed. If the

energy of the incident photons is lower and comes closer to the binding energy of the K-shell electrons, more and more of the radiation is absorbed. The highest yield is reached when the energy of the photon is just above the binding energy of the electron to be expelled. The Siegbahn notation is used to indicate which shell the electron comes from that fills the hole. In the IUPAC notation, a line is indicated by the element and the shell where the initial hole was, followed by the shell where the electron comes from that fills this hole[3]

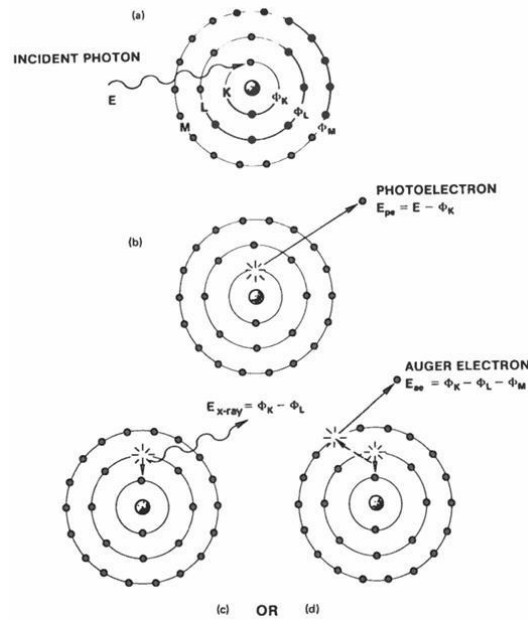


Fig 1.4 (a) Photon of energy  $E$  approaching the atom; (b) K-shell electron absorbs photon; (c) L-shell electron fills vacancy, emitting either characteristic X-rays or (d) Auger electron[6]

## XRF Classification

Basically XRF is classified mainly into two types namely Energy Dispersive X-ray Fluorescence and Wavelength Dispersive X-ray Fluorescence

### ***Energy Dispersive X-ray Fluorescence (ED-XRF)***

ED-XRF is a widely used analytical technique in which the energies of emitted fluorescent X-ray photons are directly measured using a solid-state detector, typically a Silicon Drift Detector (SDD). When a sample is irradiated with primary X-rays, inner-shell ionization occurs, leading to the emission of characteristic X-rays as electrons from higher energy levels fill the vacancies. In ED-XRF, the detector converts each incoming photon into an electrical signal proportional to its energy, thereby generating a spectrum in which peaks correspond to specific elements. This method enables rapid, simultaneous multi-element detection with relatively simple instrumentation and high count-rate capability, [2][6] ED-XRF technique is adopted for this entire thesis for its accuracy and rapid analysis.

### ***Wavelength Dispersive X-ray Fluorescence (WD-XRF)***

WD-XRF is a high-resolution analytical technique in which emitted X-rays are separated based on their wavelength using diffraction from analysing crystals, governed by Bragg's law. In this method, fluorescent X-rays emitted from the sample are directed onto a crystal monochromator, and only those satisfying the diffraction condition for a specific wavelength are detected. By scanning the angle of the crystal, different wavelengths—and hence different elemental lines—can be measured with high precision. [5][6]

## **2.6 Mechanism of XRF**

### **1. Excitation:**

- Primary X-rays (or electrons/protons) strike the sample, ejecting inner-shell electrons (e.g., K-shell) if their energy exceeds the electron binding energy.
- This creates an electron vacancy, leaving the atom in an unstable state.

### **2. Relaxation:**

- An electron from a higher-energy shell (e.g., L-shell) fills the vacancy.
- The energy difference is emitted as a characteristic X-ray photon (fluorescence) or as an Auger electron.
- The probability of X-ray emission over Auger emission is given by the fluorescence yield ( $\omega$ ), which increases with atomic number.

### **3. Detection:**

- The emitted X-rays are detected and analysed to determine elemental composition.

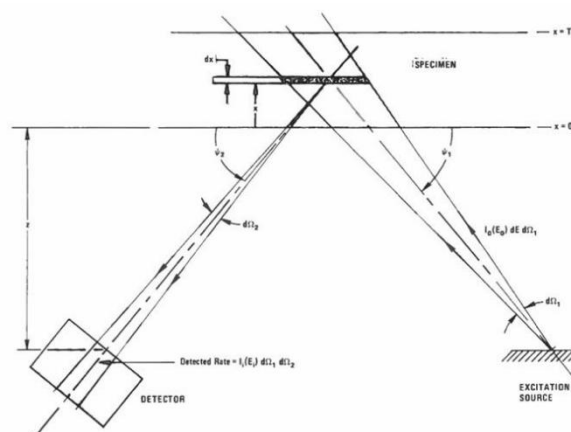


Fig 1.5 Set-up of XRF[6]

The primary fluorescence intensity ( $I_i$ ) of an element “i” is derived by considering:

- Incident X-ray flux ( $I_0(E_0)$ ): Number of photons per second at energy  $E_0$
- Absorption in the sample: Attenuation of incident and emitted X-rays.
- Photoelectric interaction: Probability of ejecting an inner-shell electron.

- Fluorescence emission: Probability of X-ray emission ( $\omega$ ) and
- detection efficiency ( $\eta$ ).[18]

## 2.7 Derivation of primary XRF intensity

1. Incident X-ray intensity reaching depth “x” in the sample:

$$I_1 = I_0(E_0) \exp[-\mu(E_0) \rho dx \csc \psi_1]$$

Where  $\mu(E_0)$  is Mass attenuation coefficient at incident energy,  $\rho$ : Density,  $\psi_1$ : Incidence angle.

2. Photoelectric interactions in a thin layer dx:

$$I_2 = I_1 \tau_i(E_0) \rho dx \csc \psi_1$$

Where  $\tau_i(E_0)$  is Photoelectric absorption coefficient for element “i”

3. K-shell ionization rate (for  $K\alpha$  emission):

$$I_3 = I_2 \left( \frac{\tau_{K_i}(E_0)}{\tau_i(E_0)} \right) = I_1 \tau_{K_i}(E_0) \rho dx \csc \psi_1$$

Where  $\tau_{K_i}(E_0)$ : K-shell absorption coefficient.

4. Fluorescence X-ray emission rate:

$$I_4 = I_3 \omega_{K_i} f$$

Where  $\omega_{K_i}$ : K-shell fluorescence yield and f: Fraction of  $K\alpha$  in the K-series.

5. Emission toward the detector (solid angle  $d\Omega_2$ ):

$$I_5 = \frac{d\Omega_2}{4\pi} I_4$$

6. Absorption of emitted X-rays (energy  $E_i$ ):

$$I_6 = I_5 \exp[-\mu(E_i) \rho x \csc \psi_2]$$

Where  $\psi_2$  - Take-off angle.

7. Detected intensity (integrating over sample thickness T and excitation energies):

$$I_i(E_i) = \frac{n(E_i)}{4\pi \sin \psi_1} \int_{E_0=E_k}^{E_{max}} Q_{if}(E_0) \times \left( \frac{1 - \exp[-\rho T(\mu(E_0) \csc \psi_1 + \mu(E_i) \csc \psi_2)]}{\mu(E_0) \csc \psi_1 + \mu(E_i) \csc \psi_2} \right) \times I_0(E_0) dE_0$$

Where  $Q_{if}(E_0) = W_i \tau_{K_i}(E_0)$  is the fluorescence efficiency and  $\eta(E_i)$ : Detector efficiency.

In addition to primary excitation, secondary and tertiary excitation effects can significantly influence the measured XRF intensity. Secondary excitation occurs when characteristic X-rays emitted by one element within the sample (e.g., element A) ionize another element (e.g.,

element B), leading to additional fluorescence signals. This effect is particularly important in multi-element samples where the emission lines of one element overlap with the absorption edges of another. Tertiary excitation, though less common, can arise when secondary X-rays further excite a third element. These enhancement effects complicate quantitative analysis but can be accounted for using correction models based on fundamental parameters.[18]

## 2.8 Factors affecting XRF

### 2.8.1 Matrix Effects

This is the most critical factor in XRF. The matrix refers to the overall chemical composition of the sample surrounding the analyte element.

- a) Absorption effect — A heavier element in the matrix absorbs the fluorescent X-rays of the analyte before they reach the detector, causing the signal to appear weaker than it truly is.
- b) Enhancement effect — The fluorescent X-rays of one element can excite another element in the sample, boosting its signal artificially.

The Sherman equation quantitatively describes the fluorescence intensity from an element  $i$  in a multi-element matrix:

$$I_i = G \cdot I_0 \cdot \frac{\mu_i^* \cdot C_i}{\bar{\mu}(E_0) + \bar{\mu}(E_i)}$$

Where  $I_i$ = measured fluorescence intensity of element  $i$ ,  $G$ = instrument geometry constant,  $I_0$ = primary beam intensity,  $\mu_i^*$ = photoelectric mass absorption coefficient of element  $i$ ,  $C_i$ = concentration of element  $i$ ,  $\bar{\mu}(E_0)$ = total matrix mass absorption at excitation energy  $E_0$ ,  $\bar{\mu}(E_i)$ = total matrix mass absorption at fluorescence energy  $E_i$

This is corrected using fundamental parameter (FP) methods or empirical correction coefficients (Lucas-Tooth and Pyne model):

$$C_i = R_i \left( 1 + \sum_j \alpha_{ij} C_j \right)$$

Where  $\alpha_{ij}$  are interelement influence coefficients.

### 2.8.2 Sample Thickness

The surface condition and homogeneity of the sample critically affect signal intensity. Key considerations:

- Surface roughness introduces geometric inconsistency — X-ray path length through the sample becomes unpredictable, increasing scatter and decreasing reproducibility.
- Particle size effects are significant for powdered samples. The intensity  $I$  varies with particle diameter  $d$  following an empirical relationship where finer grinding reduces inter-particle absorption.
- Sample thickness must exceed the infinite thickness  $t_\infty$  for complete absorption:

$$t_\infty = \frac{4.6}{\bar{\mu}(E_0) \cdot \rho}$$

Where  $\rho$  is density and  $\bar{\mu}(E_0)$  is the mass absorption coefficient at the excitation energy.

### 2.8.3 Spectral Overlap & Peak Interference

When two elements have fluorescence peaks close in energy (e.g., Mn K $\alpha$  at 5.90 keV and Fe K $\alpha$  at 6.40 keV), partial overlap occurs. The measured intensity of element  $i$  becomes:

$$I_i^{meas} = I_i^{true} + \sum_j O_{ij} \cdot I_j$$

Where  $O_{ij}$  is the overlap coefficient of element  $j$  onto element  $i$ , determined by deconvolution or reference spectra.

Pulse pile-up occurs at high count rates when two photons arrive within the detector's resolving time  $\tau$ , and are registered as a single photon of combined energy:

$$E_{pile-up} = E_1 + E_2$$

This creates ghost peaks at sum energies, visible as the dashed purple peak in the chart above.

### 2.8.4 X-ray Source Stability

The primary beam intensity  $I_0$  directly scales all fluorescence signals. Any instability introduces systematic error:

$$\frac{\Delta C_i}{C_i} \approx \frac{\Delta I_0}{I_0}$$

This is managed by:

- Ratio method — normalising analyte counts to scattered primary beam counts
- Internal standard — adding a known concentration of a reference element

- Monitor detector — a secondary detector continuously measuring  $I_0$

Tube high-voltage stability matters because the excitation efficiency  $Q(E)$  depends on overvoltage ratio  $U = V_{tube}/E_{edge}$ :

$$Q \propto \left( \frac{U - 1}{U} \right)^n, n \approx 1.67 \text{ (empirical)}$$

## 5. Geometry & Optics

The measured intensity depends strongly on the take-off angle  $\psi$  and incidence angle  $\phi$ :

$$I_i \propto \frac{\mu_i^* \cdot \csc \phi}{\bar{\mu}(E_0) \csc \phi + \bar{\mu}(E_i) \csc \psi}$$

- Shallower take-off angles increase path length, enhancing absorption and improving surface sensitivity but reducing count rate.
- Collimators and poly capillary optics define the irradiated area and solid angle subtended by the detector — both affect absolute count rate and spatial resolution.

## 6. Detector & Electronic Factors

The energy resolution  $\Delta E$  (FWHM) of an SDD is given by:

$$\Delta E = 2.355 \sqrt{F \cdot \varepsilon \cdot E + (w_{el})^2}$$

Where  $F$  = Fano factor ( $\sim 0.1$  for Si),  $\varepsilon$  = 3.6 eV per electron-hole pair,  $E$  = photon energy,  $w_{el}$  = electronic noise contribution ( $ENC \times \varepsilon$ )

Poorer resolution  $\rightarrow$  greater spectral overlap  $\rightarrow$  worse peak deconvolution. The dead time  $\tau$  of the pulse processing chain limits the maximum usable count rate:

$$N_{true} = \frac{N_{measured}}{1 - N_{measured} \cdot \tau}$$

This dead-time correction becomes critical above  $\sim 10^5$  counts/s.

# Chapter III

## 3.1 Chandrayaan II

Chandrayaan-II, India's second lunar exploration mission, was launched on 22 July 2019 by ISRO aboard a GSLV Mk-III rocket. The mission comprises an orbiter, the Vikram lander, and the Pragyan rover. While unfortunately the lander experienced a hard landing crash on 6 September 2019, the orbiter continues to function in a 100 km polar orbit around the Moon, carrying eight scientific payloads that collectively study the lunar surface, atmosphere, and environment.

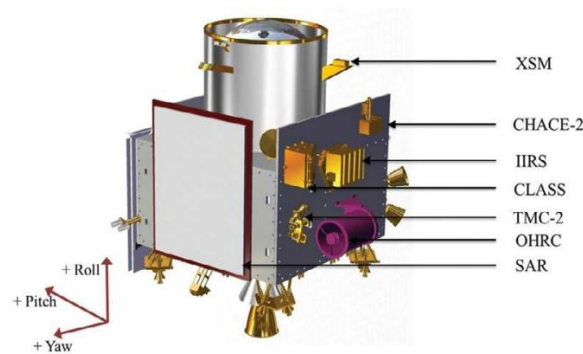


Fig 2.1 Chandrayaan II with all its payload

The Chandrayaan-2 orbiter is equipped with a suite of advanced scientific instruments designed to study the lunar surface composition, topography, mineralogy, and exosphere. Among these,

1. **CLASS** – *Chandrayaan-2 Large Area Soft X-ray Spectrometer* Measures X-ray fluorescence to determine elemental composition (Mg, Al, Si, Ca, Ti, Fe)
2. **XSM** – *Solar X-ray Monitor*  
Measures incident solar X-ray flux (important for XRF correction)
3. **CHACE-2** – *Chandra's Atmospheric Composition Explorer-2*  
Analyses lunar exosphere composition
4. **IIRS** – *Imaging Infrared Spectrometer*  
Identifies minerals and detects water/hydroxyl (OH/H<sub>2</sub>O)
5. **TMC-2** – *Terrain Mapping Camera-2*  
Provides high-resolution 3D mapping of lunar surface

6. **OHRC – Orbiter High Resolution Camera**

Captures very high-resolution images (~0.25 m resolution)

7. **DFSAR – Dual Frequency Synthetic Aperture Radar**

Studies subsurface features and detects water ice

Among these payloads, the Chandrayaan-2 Large Area Soft X-ray Spectrometer is designed to determine the elemental composition of the lunar surface using the technique of X-ray fluorescence (XRF), finding Line fluxes which is central to this study. The availability of high-resolution data from Chandrayaan-2 provides an opportunity to establish relation between geometry and line fluxes

### **3.2 Chandrayaan-2 Large Area Soft X-ray Spectrometer (CLASS)**

CLASS represents the current state of the art in space-based X-ray fluorescence spectroscopy. CLASS is a direct continuation and significant upgrade of the Chandrayaan-1 Imaging X-ray Spectrometer (C1XS), flew on Chandrayaan-1 in 2008. C1XS successfully demonstrated X-ray fluorescence mapping of the lunar surface and notably detected sodium from its characteristic X-ray line, opening up the possibility of sodium mapping on the Moon.

CLASS was designed to substantially improve upon C1XS by offering:

- Four times the detection area of C1XS
- Higher spatial resolution — 12.5 km/pixel from a 100 km orbit, compared to 25 km from C1XS
- Improved sensitivity for detecting weak signals during low solar activity
- Beryllium (Be) foils instead of aluminium (Al) foils for better low-energy X-ray transmission
- Extended energy range from 0.8 keV to 15 keV, enabling detection of Na, Mg, Al, Si, Ca, Ti, Fe and more

#### **3.2.1 Instrument Configuration**

Solar X-rays produced during a solar flare impinge on the surface of the Moon and result in fluorescent X-ray emission from the excited atoms. CLASS is designed to detect this X-ray emission in the 0.8–15 keV energy range. Since it is important to identify the spectrum of solar flare-produced X-rays that trigger the generation of fluorescent X-rays, a companion payload on Chandrayaan-2, the X-ray Solar Monitor 13, observes the Sun simultaneously to measure solar flare X-ray spectrum.

#### **3.2.2 Working Principle**

CLASS operates on the principle of X-ray fluorescence (XRF). When the Sun emits X-rays during solar flares or active periods, these X-rays irradiate the lunar surface. The atoms on the lunar surface absorb the incident X-rays and re-emit characteristic X-rays at energies unique to each element. CLASS detects these characteristic X-rays to identify and quantify the elements present on the lunar surface.

### **3.2.3 Detector System**

CLASS consists of an array of 16 swept charge devices (SCDs) operating in the energy range 0.8–15 keV (ref.14). SCDs are similar to X-ray charge coupled devices (CCDs) except for the readout architecture on the device. CLASS SCDs (CCD-236; make Teledyne e2V), have been flown on the HXMT satellite for astronomical observations 15 . Table 2 gives the device specifications. Each SCD is built with a total sensitive area of  $\sim 4 \text{ cm}^2$ , internally divided into four sections. In each section, charge is transferred towards the centre of the device, where the charge flows merge before transferring to a single charge-sensitive amplifier. Four SCDs are arranged as a quad, with the four quads forming the full sensitive array of CLASS (Figure 2). In the absence of true imaging optics, CLASS derives coarse spatial resolution using passive collimators that restrict the instantaneous field-of-view from which photons can enter the detector. CLASS collimators are designed for a ground pixel resolution of  $12.5 \text{ km} \times 12.5 \text{ km}$  on the lunar surface from a 100 km altitude orbit, which is an improvement over the C1XS capability ( $25 \text{ km} \times 25 \text{ km}$ ). The collimators made of copper coated with gold, placed above the detectors, make an opening angle of  $7.2^\circ$ , which defines the instantaneous ground pixel of  $12.5 \text{ km} \times 12.5 \text{ km}$  on the lunar surface. The SCD effective open area is 84.5% of its geometrical area.

### **3.2.4 Thermal Management**

An aluminium door protects the CLASS detectors from radiation damage during the transit from Earth to the Moon. Passive radiators connected to heat pipes provide the required low-temperature operating environment for the SCDs, which require cooling to minimize thermal noise and maintain detector performance.

### **3.2.6 Data analysis**

The output from the instrument consists of time-tagged events for each SCD, which are binned in time to generate the spectra. The spectral lines were fit using a Gaussian function and ‘mpfit’ routine in IDL, and the calibration parameters were derived for all 16

devices. The scientific data from CLASS are archived and made available in FITS format through ISRO's data archive. Each FITS file contains a primary header with mission and instrument metadata, a binary table extension listing individual photon events with their properties, and auxiliary extensions containing housekeeping and calibration data. The present project was provided with a set of CLASS FITS files by the Space Astronomy Group at URSC, Bengaluru, for the purpose of empirical calibration development.

### **3.2.7 CLASS Data products**

The high-level data products from CLASS include the following:

- Combined 8 s spectrum of all the 16 detectors in FITS format with response matrices (level 1).
- X-ray line flux from observed elements in the lunar XRF spectrum with lunar coordinates (level 2).
- Elemental maps with % weight as per coverage (level 3).

### **3.3 Role of the Companion Payload — XSM**

CLASS works in close tandem with the X-ray Solar Monitor (XSM), also onboard the Chandrayaan-2 orbiter. XSM, developed by the Physical Research Laboratory (PRL), Ahmedabad, measures the soft X-ray spectrum of the Sun in the energy range of 1–15 keV with high energy resolution ( $\sim 200$  eV at 5.9 keV). Since the solar X-ray flux drives the XRF process on the Moon, an accurate, simultaneous measurement of the solar flux is essential to convert CLASS's XRF line measurements into elemental abundances. Together, CLASS and XSM constitute the remote X-ray fluorescence spectroscopy experiment aboard Chandrayaan-2.

# Chapter IV

## 4.1 Experimental setup

The HV and LV are connected to X-ray gun through the X ray source it provides the operating voltage and tube current to produce stable X rays and the X ray gun is mounted on the motor which is rotatable and the detector will focuses on the sample and the detector is pointed normal to the sample surface which collects the XRF photons and further it sends the signal to dp5 and are analysed using suitable softwares in the PC. The below diagram shows the lab setup

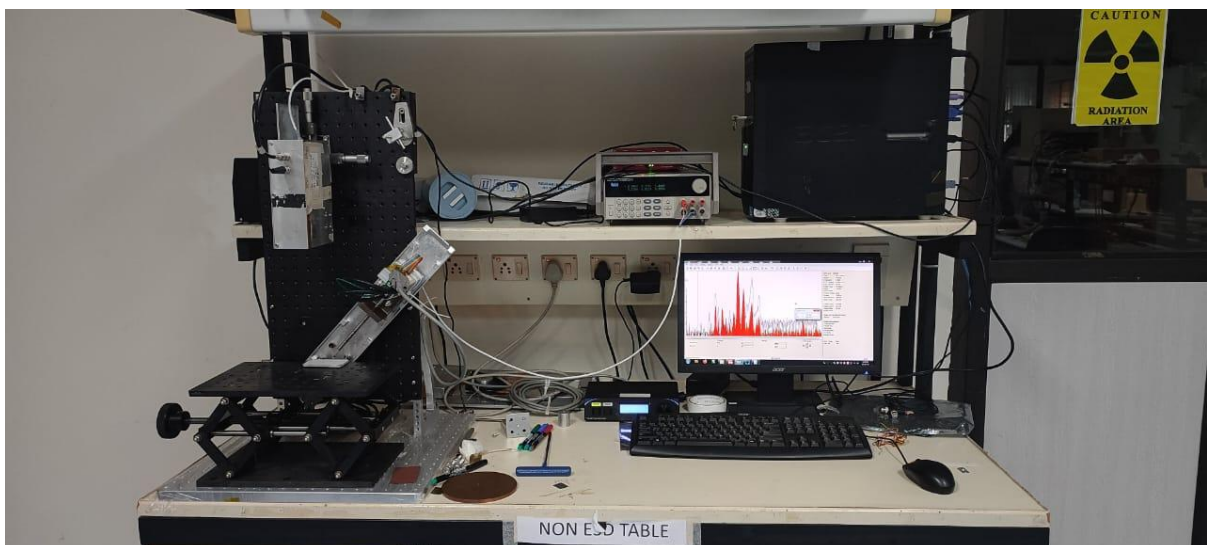


Fig 4.1: Lab set up for Angular XRF Analysis

### Silicon Drift Detector (SDD)

The Silicon Drift Detector (SDD) is a semiconductor-based radiation detector extensively used in X-ray spectroscopy due to its excellent energy resolution, high count rate capability, and low noise. The fundamental working principle of the SDD is based on the interaction of incident X-ray photons with silicon, resulting in the generation of electron-hole pairs via ionization processes. The detector consists of a series of concentric  $p^+$  electrodes biased at progressively decreasing potentials, generating a radial electric field that directs the electrons toward a small central anode [32]. This drift mechanism ensures that charge carriers travel laterally across the detector surface with minimal loss. The small size of the collecting anode significantly reduces capacitance, which is a critical factor influencing electronic noise. [33].

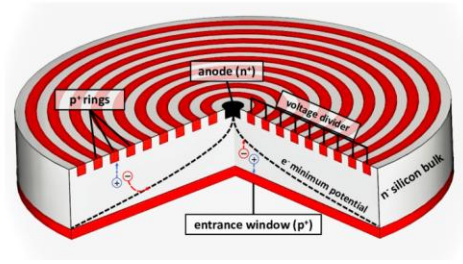


Fig Working of Silicon Drift Detector(SDD).

The energy resolution of an SDD is influenced by three primary factors: Fano noise, electronic noise, and incomplete charge collection. The Fano factor, which is approximately 0.1 for silicon, accounts for statistical fluctuations in the number of generated charge carriers and represents a fundamental limit on resolution [26]

### X-ray Gun

The X-ray gun is the primary source of X-rays in the experimental setup, where high-energy photons are generated through electron–matter interactions. It consists of a cathode (filament) and an anode (target) enclosed in a vacuum. Electrons are emitted from the heated cathode via **thermionic emission**, where thermal energy enables electrons to overcome the work function of the filament material, typically tungsten. The filament heating, controlled by the **low voltage (LV)** supply. These emitted electrons are then

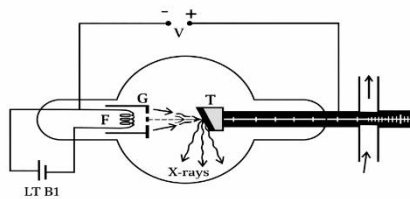


Fig Anatomy of X-ray Gun

accelerated toward the anode by the high voltage (HV) supply, acquiring kinetic energy equal to the applied potential difference

$$E = eV$$

Upon striking the target material, the electrons undergo rapid deceleration and interaction with atomic electrons, resulting in X-ray production through two main mechanisms [27], namely Bremsstrahlung Radiation, Characteristic Radiation (refer sec 2.2).

### Digital Pulse Processor

The primary function of the DPP is to perform **pulse shaping, amplification, and digitalization**. The incoming analogue signal is first shaped using digital filtering techniques to improve signal-to-noise ratio and to produce well-defined pulse heights. This shaping

process is crucial for accurate energy measurement, as the pulse height is directly proportional to the energy of the incident X-ray photon. Then DPP employs an **Analog-to-Digital Converter (ADC)** to convert each pulse into a digital value. These digital values are then sorted based on their amplitudes and stored in corresponding channels, effectively generating an energy spectrum. This process is equivalent to the function of a Multichannel Analyzer (MCA), often integrated within the DPP unit. DPP also includes features such as **pile-up rejection, dead-time correction, and baseline restoration**, which are essential for maintaining spectral accuracy

## 4.2 Experimental Procedure & Working Principle

The experimental setup was assembled as shown in Fig. X. A cylindrical collimator was attached at the exit of the X-ray tube to obtain a well-defined and narrow X-ray beam. The length of the collimator was measured to be **15.5 cm**, and its aperture diameter was **7.04 mm** measured using a screw gauge. The use of a collimator helps in reducing beam divergence and minimizes unwanted scattering, thereby improving spatial resolution and measurement accuracy.

The distance between the collimator and the sample was maintained at approximately **3 cm**, so that the primary X-ray beam remains focused on the sample surface. The detector (SDD) was positioned at a distance of **21.5 cm** from the sample, placed at normal to the sample's surface, so that it can effectively capture the emitted fluorescent X-rays while reducing the detection of scattered primary radiation additionally the X-ray tube was mounted on a motorized rotational stage to enable precise angular positioning. The motion of the stage was controlled using dedicated software (Micro Motion), allowing fine adjustment of the X-ray source orientation with a least count of **0.1°**. This arrangement provided accurate control over the incident angle of the primary X-ray beam with respect to the sample surface. Proper geometrical arrangement of the source, sample, and detector is crucial for optimizing signal quality and minimizing background noise.

The low voltage (LV) and high voltage (HV) supplies were connected to the X-ray tube. The LV supply was used to heat the filament, enabling thermionic emission of electrons, while the HV supply accelerated these electrons toward the anode target. The X-ray source was operated at **30 keV** accelerating voltage and **50  $\mu$ A** beam current. The choice of 30 keV ensures that the incident electron energy is sufficient to excite characteristic X-rays from a wide range of elements present in the samples like Fe, Cu, Ni etc., particularly by overcoming inner-shell binding energies. The beam current of 50  $\mu$ A controls the number of electrons striking the

target, thereby regulating the intensity of the generated X-rays. This combination provides a good balance between sufficient X-ray intensity and manageable thermal load on the target.

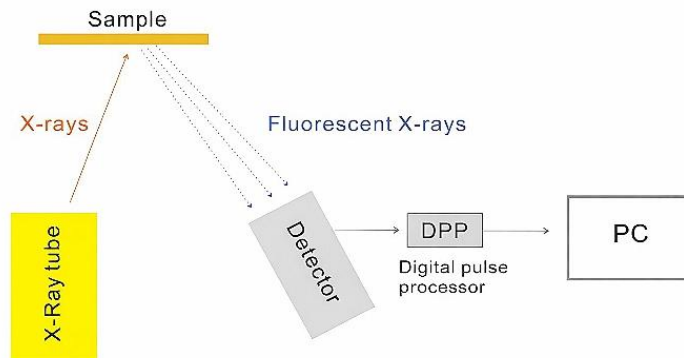


Fig 4.2 Schematic diagram represents the procedure of Working

Once the X-ray source is switched on, primary X-rays are generated and directed toward the sample through the collimator. When these incident X-rays interact with the atoms in the sample, the **X-ray fluorescence (XRF) process** occurs. In this process, primary photons eject inner-shell electrons (typically from K or L shells) via the photoelectric effect. The resulting vacancies are filled by electrons from higher energy levels, leading to the emission of secondary (fluorescent) X-rays with energies characteristic of the elements present in the sample. The emitted fluorescent X-rays are detected by the **Silicon Drift Detector (SDD)**, which converts the incoming photons into electrical charge signals proportional to their energy. These signals are then passed to the **Digital Pulse Processor (DPP)**, where they undergo pulse shaping, amplification, and analogue-to-digital conversion. The processed digital signals are sorted into energy channels to form a spectrum. Finally, the generated spectrum is displayed on the computer using appropriate data acquisition software. The peaks observed in the spectrum correspond to characteristic X-ray energies of elements in the sample, enabling qualitative and quantitative analysis.

### 4.3 Beam Geometry

The X-ray gun was fitted with a cylindrical collimator of aperture diameter  $D = 7.04$  mm and length  $L = 155$  mm. The collimator-to-sample distance was approximately 30 mm. The half-angle of beam divergence is:

$$\theta_{div} = \tan^{-1} \left( \frac{D}{2L} \right) = \tan^{-1} \left( \frac{7.04}{310} \right) \approx 1.30^\circ$$

When the divergent beam is incident on the sample at an angle ( $\theta$ ), the initially circular beam spot transforms into an elliptical footprint due to geometric projection.

- **Minor axis of ellipse:**

$$b = D$$

- **Major axis of ellipse:**  $a = \frac{D}{\sin \theta}$

where  $D$  is the beam diameter.

The area of the elliptical footprint is given by:

$$A(\theta) = \pi \cdot \frac{a}{2} \cdot \frac{b}{2}$$

This calculation was performed for incidence angles ranging from **5° to 70°** in steps of 5°, resulting in a systematic variation of illuminated area with angle.

### **Normalization of Area**

To facilitate comparison, all calculated footprint areas were normalized with respect to the maximum area (corresponding to the lowest incidence angle):

$$Area_{norm}(\theta) = \frac{Area(\theta)}{Area_{max}}$$

This normalization converts the absolute area into a dimensionless scaling factor, representing the relative geometric spreading of the beam.

### **Area Correction of Line Flux**

The measured line flux (counts) is influenced by the illuminated area due to geometric spreading of the beam. At lower incidence angles, the footprint area is larger, leading to reduced counts per unit area, while at higher angles the footprint decreases, concentrating the flux.

To remove this geometric effect, the measured counts were corrected using the footprint area:

$$F_{corrected}(\theta) = \frac{F_{measured}(\theta)}{Area(\theta)}$$

This correction yields the **intensity per unit area**, effectively isolating the intrinsic dependence of line flux on incidence angle.

## 4.4 Samples Used

Seven samples were characterised under the same condition spanning from pure metals, alloys, compound semiconductor to planetary regolith simulant they are as below

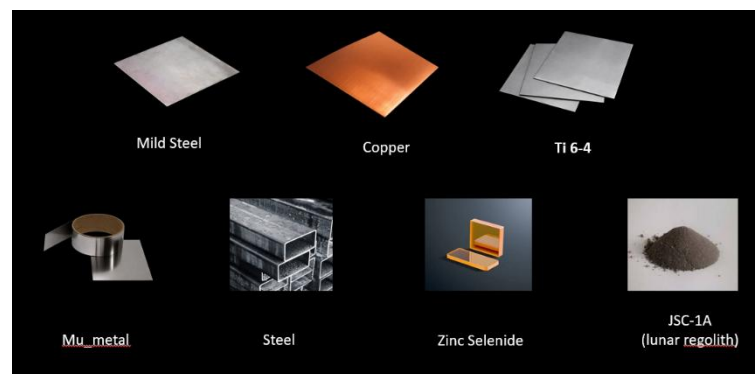


Fig 4.3 images of samples used

## 4.5 Experimental Conditions and Precautions

The experiments were carried out under **atmospheric conditions**, without the use of a vacuum or helium environment. Due to the presence of air, attenuation of low-energy X-rays is significant, particularly for elements such as **Mg, Al, Si** etc. As a result, characteristic X-rays from light elements such as Mg, Al, and Si are more susceptible to absorption before reaching the detector. On the other-hand elements like **Ca, Ti, V, Cr, Mn, Fe, Cu, Zn, Se** shows high response.

To ensure stable operation of the instrumentation, the laboratory temperature was maintained at a relatively low and controlled level throughout the experiment. This is essential because both the X-ray tube and the detector system are sensitive to temperature variations. Elevated temperatures can introduce electronic noise, drift in detector response, and instability in measurement conditions. Maintaining a controlled environment helps improve spectral stability and reproducibility.

Appropriate radiation safety precautions were strictly followed during the experiment. Since X-rays are ionizing in nature and can pose health hazards, access to the laboratory was restricted during operation of the X-ray source. No personnel were allowed inside the experimental area while the X-ray beam was active, ensuring safe working conditions.

All experiments and measurements were carried out at the **Instrumentation Laboratory of the Space Science Instrumentation Facility (SSIF), Space Astronomy Group (SAG), U R Rao Satellite Centre (URSC), ISRO, Bangalore**. The use of this well-equipped and controlled laboratory environment ensured the reliability and accuracy of the experimental results.

## 4.6 Data Acquisition Procedure

### 4.6.1 Measurement Protocol

The measurement sequence for each sample was as follows:

- The sample was placed on the vertically movable stage beneath the detector.
- The X-ray gun was positioned at  $10^\circ$  with respect to the sample surface and powered on.
- A spectrum was acquired over 300 s using a Multi-Channel Analyser (MCA) connected to the detector. The MCA binned the incoming photon pulse heights into discrete channels.
- The gun was stepped through  $15^\circ$ ,  $20^\circ$ ,  $25^\circ$ ,  $30^\circ$ ,  $35^\circ$ ,  $40^\circ$ ,  $45^\circ$ ,  $50^\circ$ ,  $55^\circ$ ,  $60^\circ$ ,  $65^\circ$ , and  $70^\circ$ , with a 300 s acquisition at each position.
- The procedure was repeated for all six samples, maintaining identical geometric and electrical conditions.

### 4.6.2 MCA and Channel Number Assignment

The MCA stores a raw pulse-height spectrum. Each ADC channel corresponds to a voltage bin from the detector pre-amplifier output. Since the spectra files stored by the MCA contained only photon counts per bin (with no channel index column), a Python script was written to:

- Read the raw counts array from each file.
- Generate a channel number array [0, 1, 2, ..., N-1] indexed from zero.
- Produce an initial channel-number vs. counts plot (the 'raw spectrum').

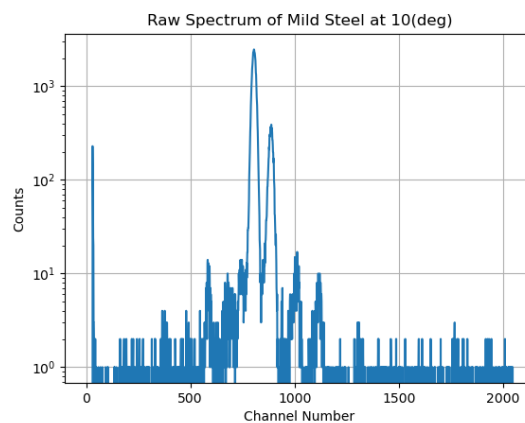


Fig 4.4: Raw spectrum of mild steel taken at 10 degree angle at log scale in y axis

At this stage, peaks are visible as functions of channel number, but their physical energy identity is unknown without calibration.

## 4.7 Peak Identification and Gaussian Fitting

### 4.7.1 Initial Spectral Inspection

Examination of the mild steel raw spectrum revealed two prominent peaks: a strong peak at channel 804 and a weaker peak at channel 886. In XRF, iron (Fe) is the dominant element in mild steel, so these were immediately hypothesised to correspond to the Fe  $K\alpha$  (6.404 keV) and Fe  $K\beta$  (7.058 keV) emission lines respectively, which are the two most intense characteristic X-ray lines of iron in this energy regime.

### 4.7.2 Physical Origin of Fe $K\alpha$ and $K\beta$ Lines

When X-rays from the tube (with energies above 7.058 keV, the Fe K-shell binding energy) irradiate the steel, they photo-ionise K-shell electrons from iron atoms. The resulting K-shell vacancy is filled by electron transitions from higher shells:

- Fe  $K\alpha$ : L-shell  $\rightarrow$  K-shell transition. This produces the  $K\alpha_1$  (6.404 keV) and  $K\alpha_2$  (6.391 keV) doublet, which are unresolved by most solid-state detectors and by using the below formula

$$K\alpha = \frac{(\text{rad rate of } K\alpha_1) \times E(K\alpha_1) + (\text{rad rate of } K\alpha_2) \times E(K\alpha_2)}{\text{rad rate of } K\alpha_1 + \text{rad rate of } K\alpha_2}$$

The values are taken from the X-raylib datas and resolved as a single peak at  $\approx 6.399$  keV.

- Fe  $K\beta$ : M-shell  $\rightarrow$  K-shell transition, producing the  $K\beta$  line at  $\approx 7.058$  keV. This line is less intense than  $K\alpha$  (approximately 13–15% of  $K\alpha$  intensity) due to lower transition probability.

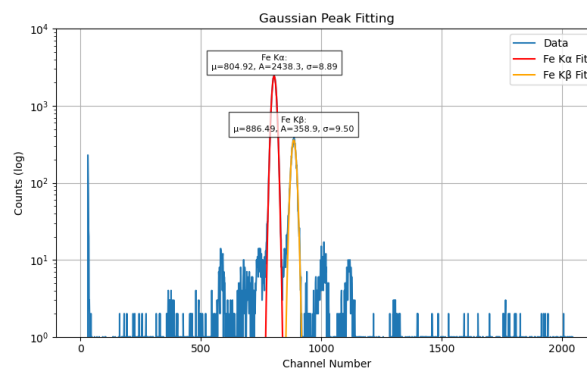


Fig 4.5 Gaussian fitted peak with parameters – mild steel

### 4.7.3 Gaussian Fitting

The Gaussian function is defined by three key parameters: **amplitude (A)** represents the peak height, **mean ( $\mu$ )** corresponds to the peak centre; and **standard deviation ( $\sigma$ )** which indicates the peak width. By fitting the experimental data with this function. The raw spectral data, which contains statistical fluctuations and noise, was fitted with a Gaussian function to obtain precise peak characteristics

Each identified peak was fitted with a Gaussian profile of the form:

$$N(ch) = A \exp\left(-\frac{(ch - ch_0)^2}{2\sigma^2}\right)$$

Where

$A$  is the peak amplitude (counts),

$ch_0$  or  $\mu$  is the centroid channel,

$\sigma$  is the peak width.

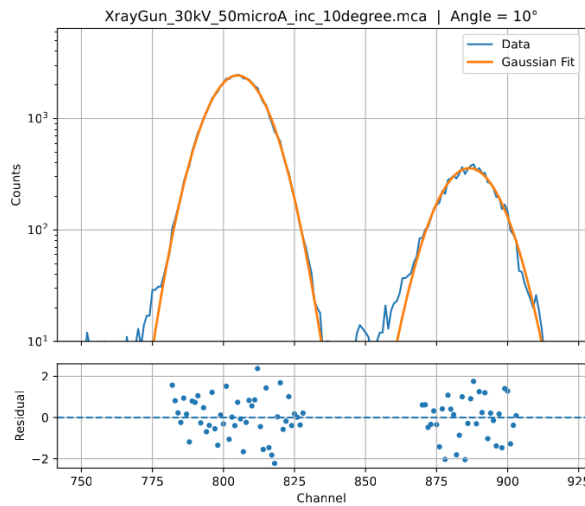


Fig 4.6 mild steel gaussian fitted curves with residuals

residual plot (difference between observed data and fitted model) shows random distribution around zero, confirming the reliability and quality of the fit. This indicates that the Gaussian model adequately describes the peak shape without significant systematic deviation

Gaussian fitting is appropriate here because:

- Semiconductor detectors (e.g. Si-PIN or Si-drift detectors) produce a Gaussian peak profile due to statistical charge collection fluctuations.

- The FWHM of the Gaussian ( $= 2.355\sigma$  in energy units) quantifies the detector energy resolution, typically  $\sim 150\text{--}250$  eV at 5.9 keV for an Si-PIN detector.



Fig 4.7 Stability of the Fe K $\alpha$  peak Mild steel

The above plot evaluates the stability of the Fe K $\alpha$  peak positions across all incident angles ranges from  $10^\circ$  to  $70^\circ$ . This analysis focuses on the Fe K $\alpha$  peak due to its high intensity (norm), making it a reliable indicator of detector performance over time. The plot tracks the fitted Gaussian centre ( $\mu$ ) of the Fe K $\alpha$  peak for spectrum at each angle on a  $5^\circ$  gap, providing insights into the consistency of the measured peaks.

## 4.8 Energy Calibration:

### 4.8.1 Channel-to-Energy Conversion

Assuming a linear relationship between channel number (C) and photon energy (E), the calibration was carried out using the equation

$$E = mC + c$$

where  $m$  represents the gain (slope) of the system and  $c$  represents the offset. Using the known energy–channel pairs, the calibration constants were determined through linear fitting

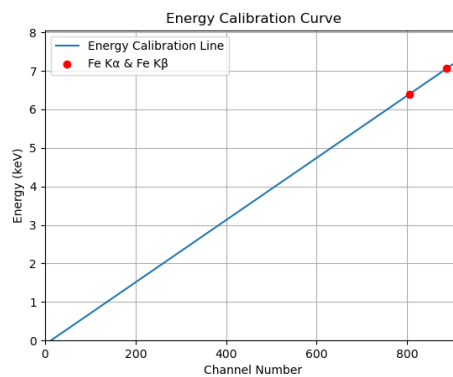


Fig 4.8 Energy Calibration Graph

The relationship between MCA channel number and photon energy is the excellent approximation, linear, for a well-behaved detector-amplifier chain the **Gain** is the energy per channel (keV/channel)

and **Offset** accounts for the zero-energy intercept of the ADC (electronic offset). These two parameters fully characterise the energy scale of the MCA.

## 4.8.2 Determining Gain and Offset

With two known energy-channel pairs from the Fe spectrum:

- Fe K $\alpha$ :  $E_1 = 6.399$  keV,  $ch_1 = 804.49$
- Fe K $\beta$ :  $E_2 = 7.058$  keV,  $ch_2 = 886.92$

The calibration equations are:

$$6.399 = \text{Gain} \times 804.92 + \text{Offset}$$

$$7.058 = \text{Gain} \times 886.49 + \text{Offset}$$

Subtracting the first from the second:

$$7.058 - 6.399 = \text{Gain} \times (886.49 - 804.92)$$

$$0.659 = \text{Gain} \times 81.57$$

$$\text{Gain} = 0.659 / 81.57 \approx 0.0080789505 \text{ keV/channel} \approx 8.0789 \text{ eV/channel}$$

Substituting back to find Offset:

$$\text{Offset} = 6.399 - (0.0080789505 \times 804.92) = 6.399 - 6.502 \approx -0.103 \text{ keV}$$

The near-zero offset confirms that the ADC electronics are well-calibrated with essentially no electronic zero-shift.

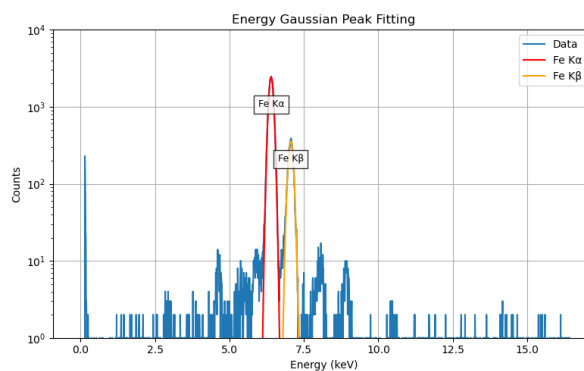


Fig 4.9 Mild Steel energy fitted Gaussian Curve

## 4.9. Flux Determination

### 4.9.1 Line Flux

The area under each Gaussian peak was calculated to determine the line flux of the characteristic X-rays. The line flux is given by:

$$\text{Line flux} \propto A \cdot \sigma \cdot \sqrt{2\pi}$$

where  $A$  is the amplitude and  $\sigma$  is the standard deviation of the Gaussian. This quantity represents the total number of detected photons corresponding to a specific transition ( $K\alpha$  or  $K\beta$ ).

The calculated line flux values provide a quantitative measure of the emitted X-ray intensity and are essential for further analysis. Line flux (area under a Gaussian peak) directly represents the total number of fluorescence photons detected in that emission line during the acquisition time. For a Gaussian with fitted parameters ( $A$ ,  $\text{cho}$ ,  $\sigma$ ) This area has units of **counts**. Dividing by the acquisition time ( $T = 300$  s) converts it to a count rate (counts per second) represents the **Line flux intensity or normalised line flux (cps)** which is proportional to the fluorescence photon flux incident on the detector

$$\text{Normalised Line Flux (cps)} = \frac{\text{Line Flux (counts)}}{W \text{ (s)}}$$

### 4.9.2 Poisson Statistics for error analysis

X-ray detection is a counting process in which each photon arrives independently and at random. The number of counts in a given time interval follows a Poisson distribution. For a Poisson process with mean  $\mu$  counts, the standard deviation is:

$$\sigma_{\text{counts}} = \sqrt{\mu}$$

Therefore, the statistical uncertainty ( $1\sigma$ ) in the measured peak area  $N$  is simply:

$$\delta N = \sqrt{N}$$

This gives a **relative uncertainty** of  $1/\sqrt{N}$ . For a peak area of  $N = 10,000$  counts:  $\delta N = 100$  counts, i.e. 1% relative precision. This is why long acquisition times are beneficial — they reduce relative statistical noise.

The uncertainty in the count rate  $R = N/T$  is propagated as:

$$\delta R = \delta N / T = \sqrt{N} / T$$

Here, the corrected signal intensity ( $S_{\text{Area}}$ ) is obtained by normalizing the measured Gaussian peak area ( $\text{Area}$ ) with respect to the effective beam spot area ( $\text{Spot\_area}$ ), such that

$$S_{\text{Area}} = \frac{\text{Area}}{\text{Spot\_area}}$$

This normalization is essential to remove geometric effects arising from beam divergence and variation in incidence angle. Since (*Sarea*) is a ratio of two independently measured quantities, its uncertainty depends on the uncertainties in both (*Area*) and (*Spot\_area*). For independent variables, the propagated uncertainty is given by:

$$\Delta SArea = SArea \times \sqrt{\left(\frac{\Delta Area}{Area}\right)^2 + \left(\frac{\Delta Spot\_area}{Spot\_area}\right)^2}$$

Here,  $\left(\frac{\Delta Area}{Area}\right)$  and  $\left(\frac{\Delta Spot\_area}{Spot\_area}\right)$  represent the relative (fractional) uncertainties. The uncertainty in (*Area*) is typically governed by counting statistics, where photon detection follows Poisson behavior; hence,  $(\Delta Area \approx \sqrt{Area})$ . The uncertainty in (*Spot\_area*) arises from experimental factors such as beam divergence, alignment errors, and angular resolution,

## 4.10 Applying to all Samples

Energy calibration of the detector system was carried out using **mild steel** as a reference material here. Mild steel was chosen due to the presence of well-defined characteristic X-ray peaks (primarily Fe K $\alpha$  and K $\beta$  lines), which serve as reliable reference points for calibration. Before the measurement of each sample, a spectrum of the mild steel reference was recorded under identical experimental conditions. Similarly, after completing the measurement of the sample, the reference spectrum was acquired again. This pre- and post-measurement calibration approach was adopted to account for any possible drift in detector response, electronic fluctuations, or environmental variations during the experiment. The peak positions corresponding to the characteristic energies of iron were identified in both spectra. The average of these peak positions was then calculated and used to perform energy calibration for the corresponding sample measurement. This averaging method improves the accuracy and stability of calibration by minimizing systematic errors and temporal variations in the system. The use of repeated reference measurements and consistent operating conditions enhances the precision and reliability of the experimental results.

## Mu Metal

**Mu-metal** is a nickel–iron based soft ferromagnetic alloy known for its exceptionally high magnetic permeability. It typically consists of approximately 75–80% nickel (Ni), 15–20% iron (Fe), along with small amounts of Manganese (Mn). Due to its unique magnetic properties, mu-metal is widely used in magnetic shielding applications

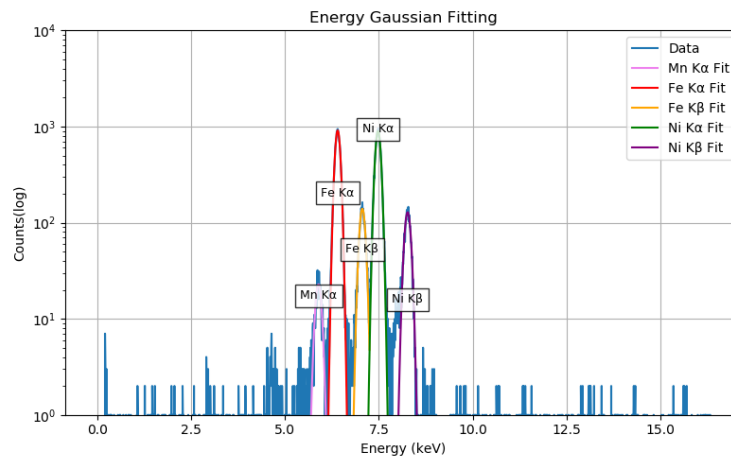


Fig 4.10 Mu metal energy fitted Gaussian Curve

After energy channel calibration the elements in the samples are found and classified by the XRF technique under lab setup mentioned above.

## Steel

The Steel sample is a complex multi-element alloy with high iron content, nominally comprising approximately 65–70 wt.% Fe, 18–20 wt.% Cr, 8–12 wt.% Ni, and ~2 wt.% Mn (consistent with AISI 304-type stainless steel)

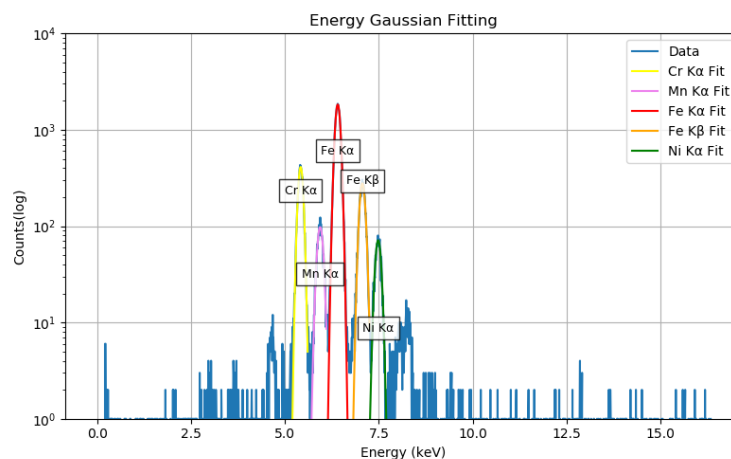


Fig 4.11 Steel energy fitted Gaussian Curve

The above figure confirms that after energy channel calibration the elements or lines in the sample steel is confirmed as Cr K $\alpha$ , Mn K $\alpha$ , Fe K $\alpha$ , Fe K $\beta$ , Ni K $\alpha$

## Copper

**Copper (Cu)** is a pure metallic element with atomic number 29, widely known for its excellent electrical and thermal conductivity. In X-ray fluorescence (XRF) analysis, copper serves as an ideal sample due to its well-defined and intense characteristic X-ray emission lines

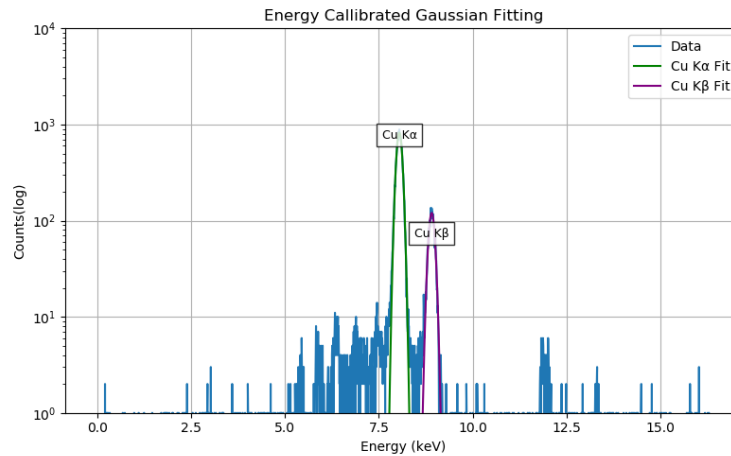


Fig 4.12 Copper energy fitted Gaussian Curve

After energy channel calibration the elements in the samples are found and classified by the XRF technique under lab setup mentioned above. The figure confirms that Copper consists of Cu K $\alpha$ , Cu K $\beta$

## Zinc Selenide

**Zinc selenide (ZnSe) is a lemon-yellow, crystalline intrinsic semiconductor.** It is widely used in optical and detector applications due to its wide bandgap and good transmission in the infrared region. Unlike metallic samples, ZnSe exhibits semiconductor behaviour. The K-shell binding energies are Zn K $\alpha$  = , Zn K $\beta$  = , Se K $\alpha$  = , Se K $\beta$  = respectively.

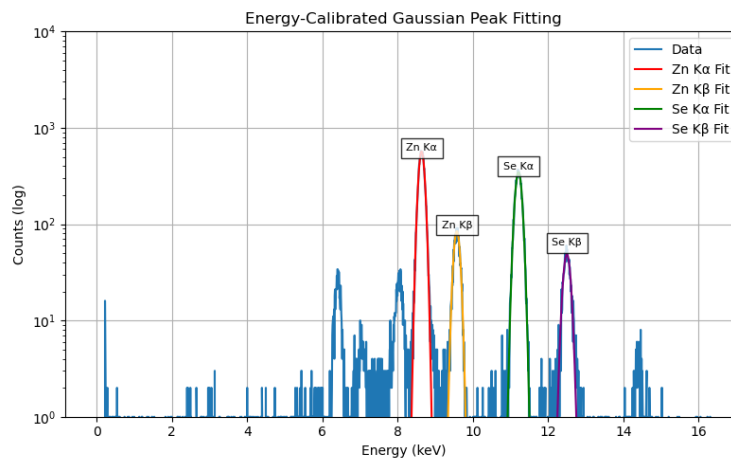


Fig 4.13 Zinc Selenide energy fitted Gaussian Curve

After energy channel calibration the elements in the sample ZnSe consists of Zn K $\alpha$ , Zn K $\beta$ , Se K $\alpha$ , Se K $\beta$

## Ti-6-4

Ti-6Al-4V (Ti64) is a titanium-based alloy composed of approximately 90% titanium (Ti), 6% aluminium (Al), and 4% vanadium (V) by weight. It is one of the most widely used titanium

alloys due to its high strength-to-weight ratio, excellent corrosion resistance and good mechanical properties

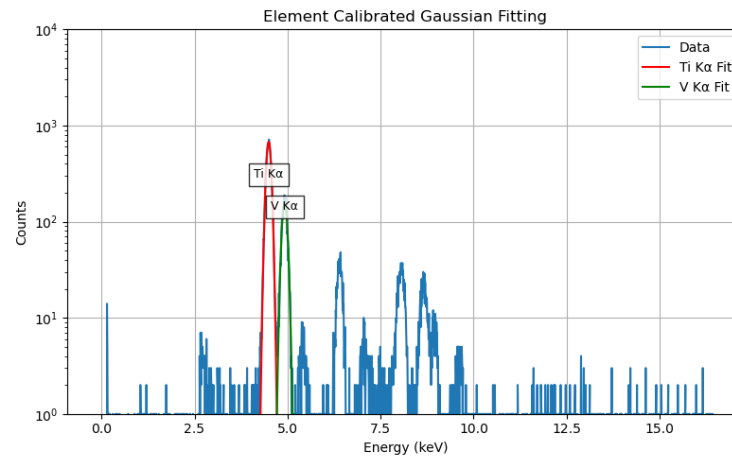


Fig 4.14 Ti-6-4 energy fitted Gaussian Curve

After energy channel calibration the element's line in the samples are found as Ti K $\alpha$ , V K $\alpha$ .

## JSC-1A Lunar Regolith

**JSC-1A** is a lunar regolith simulant developed on Earth to mimic the composition and properties of Moon. JSC-1A lunar regolith simulant consists of multiple elements including approximately **20–22% silicon (Si)**, **6–8% aluminium (Al)**, **8–12% iron (Fe)**, **6–8% calcium (Ca)**, **4–6% magnesium (Mg)**, and **1–3% titanium (Ti)**, along with trace amounts of sodium (Na), potassium (K), and manganese (Mn). The applied 30KeV is enough to kick out electron from the k shell

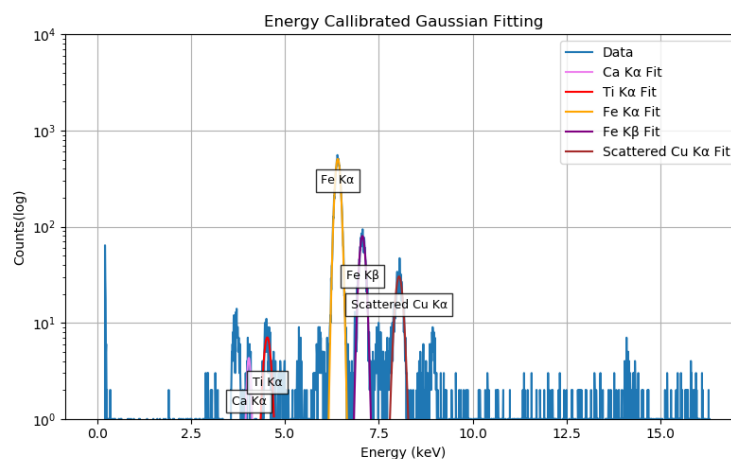


Fig 4.15 JSC-1A energy fitted Gaussian Curve

After energy channel calibration the elements in the samples are found as Ca K $\alpha$ , Ti K $\alpha$ , Fe K $\alpha$ , Fe K $\beta$ , Scattered Cu K $\alpha$ .

Note: All the samples were examined at same condition as mentioned above (ref sec 4.7).

## 4.11 Angular dependence of line flux

All measured elemental XRF emission lines, including their  $K\alpha$  and  $K\beta$  components, were utilized to generate their line flux(counts) (ref sec 4.11.1) versus angle distributions are listed below. Error bars(ref sec 4.11.2) are also overlaid on the plot and are computed based on Poisson uncertainties propagated through the count ratios. The same analysis methodology was applied to all the samples to ensure consistency in the results and for creating the empirical relations

### Mild Steel

The line flux obtained from Fe  $K\alpha$  and Fe  $K\beta$  emissions in mild steel has been extracted and plotted as a function of angle.

#### *Iron $K\alpha$ & $K\beta$*

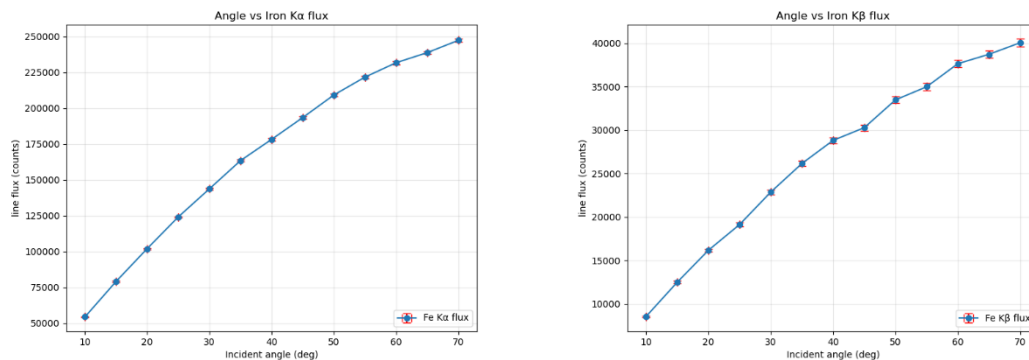


Figure 3.3: Incident angle vs Fe  $K\alpha$  &  $K\beta$  line flux

### Mu Metal

The line flux obtained from Mn  $K\alpha$ , Fe  $K\alpha$ , Fe  $K\beta$ , Ni  $K\alpha$ , Ni  $K\beta$  emissions in Mu-metal has been extracted and plotted as a function of angle.

## Manganese Ka

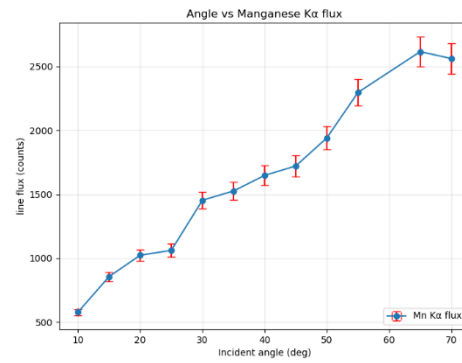


Figure 3.7: Incident angle vs Mn Ka line.

## Iron Ka & K $\beta$

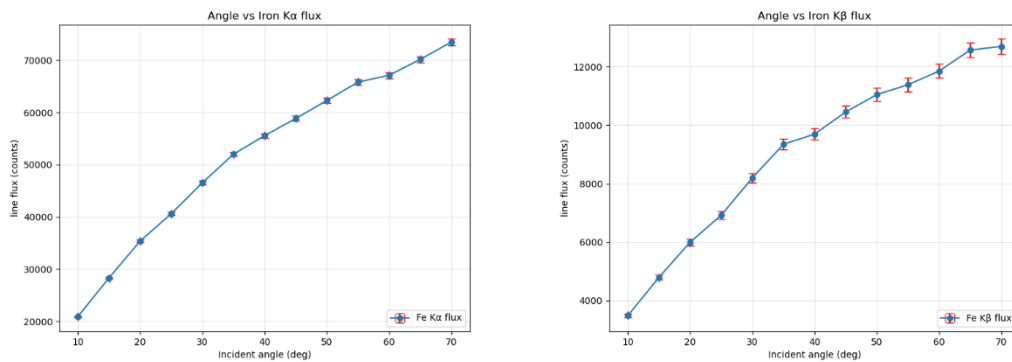


Figure 3.8: Incident angle vs Fe Ka & Fe K $\beta$  line flux(counts)

## Nickel Ka & K $\beta$

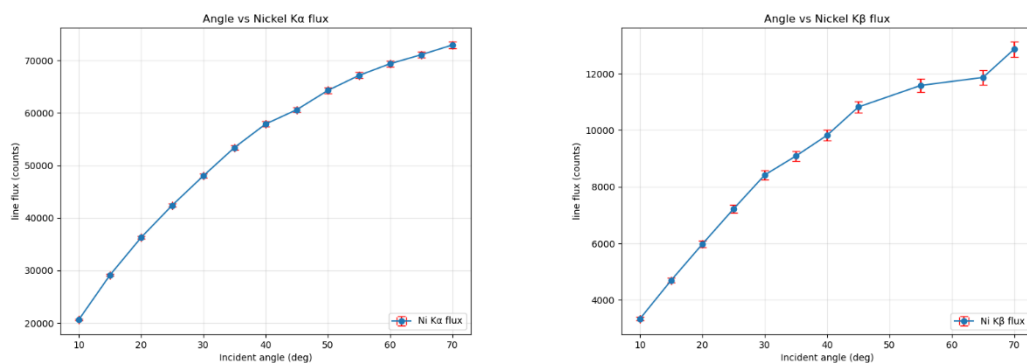


Figure 3.6: Incident angle vs Ni Ka & Ni K $\beta$  line flux (counts)

## Steel

The line flux obtained from Cr K $\alpha$ , Mn K $\alpha$ , Fe K $\alpha$ , Fe K $\beta$ , Ni K $\alpha$  emissions in Mu-metal has been extracted and plotted as a function of angle

### *Chromium K $\alpha$ & Manganese K $\alpha$*

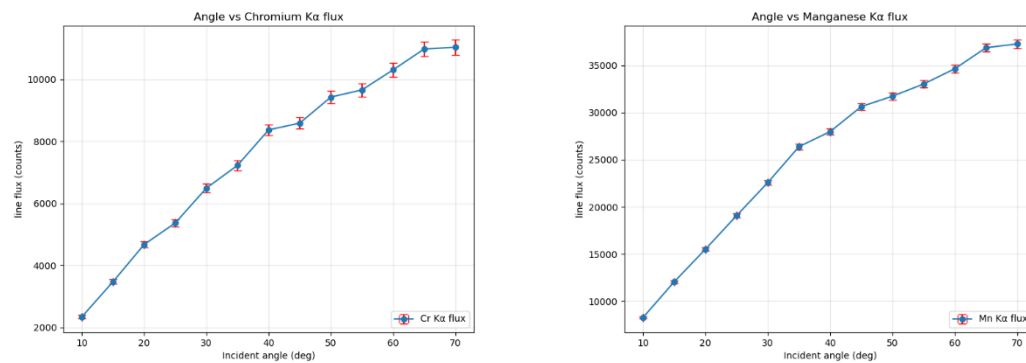


Figure 3.18: incident angle vs Cr K $\alpha$  & Mn line flux (counts)

### *Iron K $\alpha$ & Iron K $\beta$*

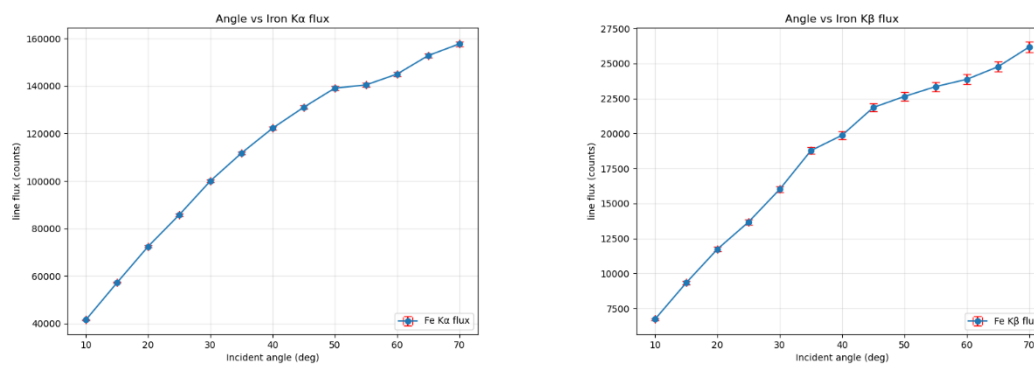


Figure 3.17: Incident angle vs Fe K $\alpha$  & K $\beta$  line flux (counts)

## Nickel $K\alpha$

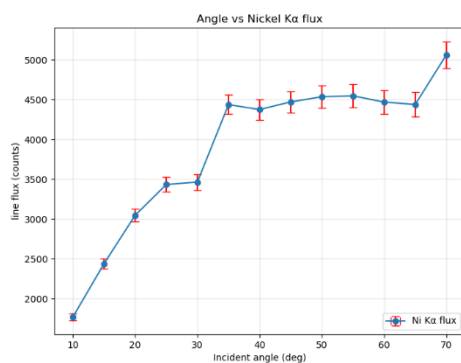


Figure 3.14: Incident angle vs Ni  $K\alpha$  line flux

## Copper

The line flux obtained from Cu  $K\alpha$  and Cu  $K\beta$  emissions in mild steel has been extracted and plotted as a function of angle

### Copper $K\alpha$ & $K\beta$

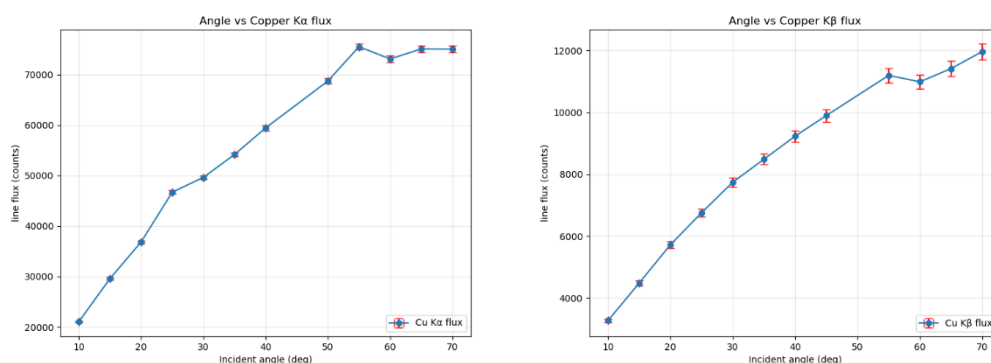


Figure 3.11: Incident angle vs Cu  $K\alpha$  and Cu  $K\beta$  line flux (counts)

## Zinc Selenide (ZnSe)

The line flux obtained from Zn  $K\alpha$ , Zn  $K\beta$ , Se  $K\alpha$ , Se  $K\beta$  emissions in Mu-metal has been extracted and plotted as a function of angle

### Zinc $K\alpha$ & $K\beta$

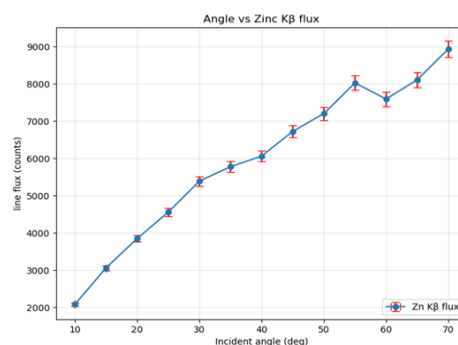
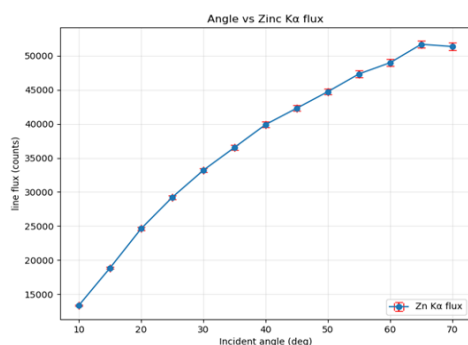


Figure 3.13: Incident Angle vs Zn  $K\alpha$  and Zn  $K\beta$  line flux

## Selenium $K\alpha$ & $K\beta$

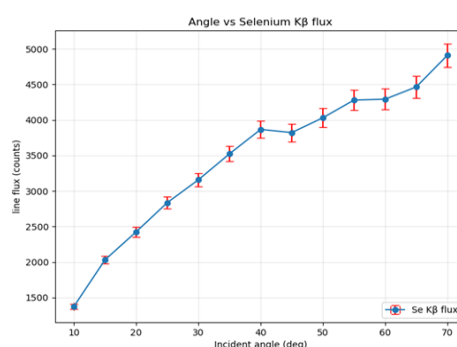
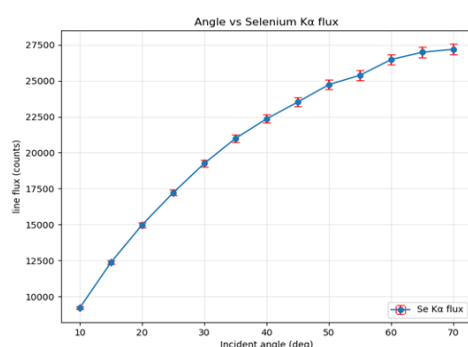


Figure 3.15: Incident angle vs Se  $K\alpha$  &  $K\beta$  line flux (counts)

## Ti-6Al-4V (Ti64)

The line flux obtained from Ti  $K\alpha$ , V  $K\alpha$  emissions in Mu-metal has been extracted and plotted as a function of angle

## Titanium $K\alpha$ & Vanadium $K\alpha$

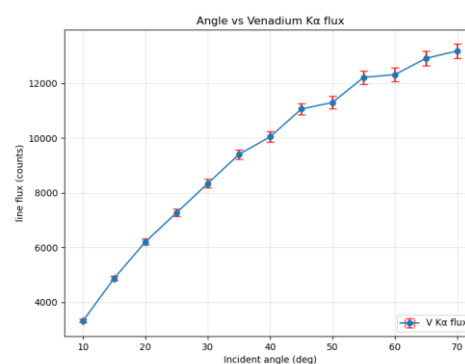
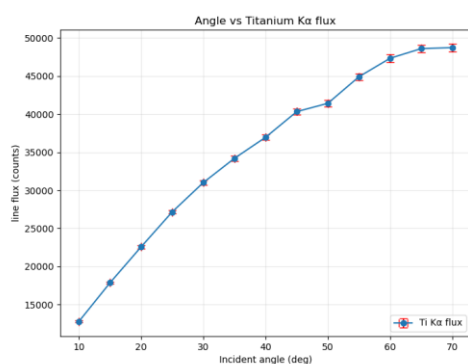


Figure 3.17: Incident angle vs Ti  $K\alpha$  and V  $K\alpha$  line flux.

## JSC – 1A

The line flux obtained from Ca K $\alpha$ , Ti K $\alpha$ , Fe K $\alpha$ , Fe K $\beta$  and scattered Cu K $\alpha$  emissions in Mu-metal has been extracted and plotted as a function of angle

### *Calcium K $\alpha$ & Titanium K $\alpha$*

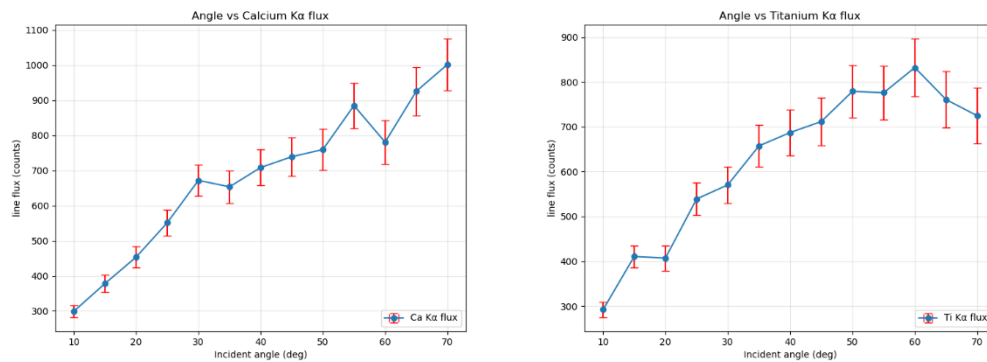


Figure 3.15: Incident angle vs Ca K $\alpha$  & Ti K $\alpha$  line flux (counts)

### *Iron K $\alpha$*

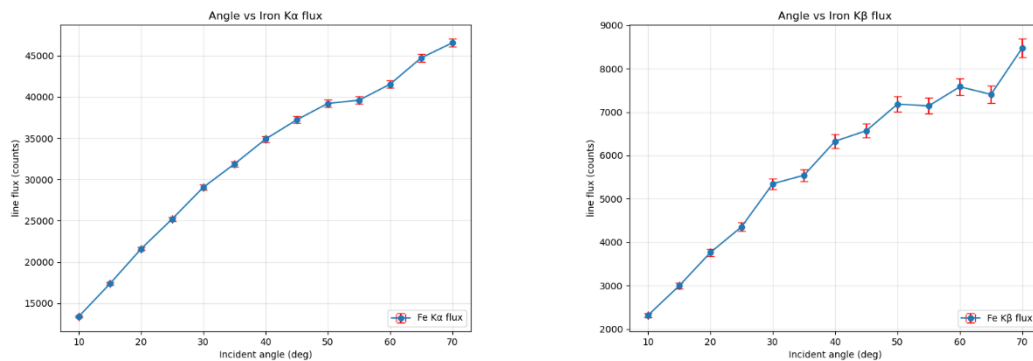


Figure 3.21: Incident Angle vs Fe K $\alpha$  & K $\beta$  line flux (counts)

## Scattered Copper Ka

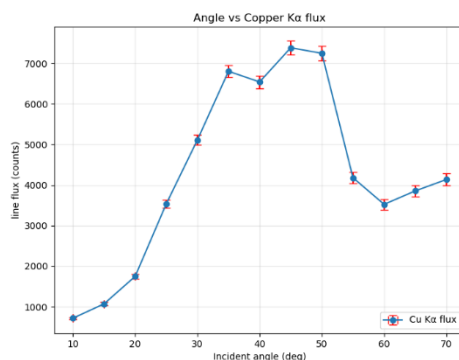


Figure 3.21: Incident Angle vs Scattered Cu Ka line flux

The variation of line flux with incident angle was studied by plotting line flux (counts) as a function of angle. The graphs shows a clear increasing trend in flux with increasing incident angle.

This behaviour can be physically explained based on the geometry of interaction:

- At lower angles, the effective path length of X-rays within the sample is small, leading to reduced interaction volume and lower fluorescence yield.
- As the incident angle increases, the interaction volume within the sample increases, resulting in higher probability of photoelectric absorption and hence greater fluorescence emission.
- Additionally, the escape probability of fluorescent X-rays toward the detector improves at higher angles, contributing to increased detected intensity.

The divergence of the beam, previously determined, was incorporated into the analysis to ensure accurate results

# Chapter V

The scientific data from CLASS are archived and made available in FITS format through ISRO's data archive. Each FITS file contains a primary header with mission and instrument metadata, a binary table extension listing individual photon events with their properties, and auxiliary extensions containing housekeeping and calibration data. The present project was provided with a set of CLASS FITS files by the Space Astronomy Group at URSC, Bengaluru, for the purpose of empirical calibration development.

## 5.1 Solar Flare

Solar flares are the main source of X-rays in the Space. They are the sudden and intense releases of energy that occur in the Sun's atmosphere, primarily in active regions associated with sunspots. These events are among the most energetic phenomena in the solar system and are characterized by the rapid emission of electromagnetic radiation across a wide spectrum, including radio waves, visible light, ultraviolet radiation, X-rays, and gamma rays.

The primary mechanism responsible for solar flares is known as Magnetic Reconnection. In active regions of the Sun, magnetic field lines become highly twisted and stressed due to differential rotation and plasma motion. When oppositely directed magnetic field lines come into close proximity, they undergo reconnection, resulting in a sudden release of stored magnetic energy. This energy is rapidly converted into thermal energy, kinetic energy of accelerated particles, and electromagnetic radiation. As a result, plasma in the solar atmosphere can reach temperatures of the order of  $10^7$  K, and high-energy electrons and ions are accelerated. Solar flares produce a significant flux of high-energy X-rays, which are particularly important for inducing X-ray Fluorescence in planetary surfaces

Solar flares are classified as below.

| Flare Type | X-ray Flux (W/m <sup>2</sup> ) | Energy Range (keV) |
|------------|--------------------------------|--------------------|
| A-class    | $< 10^{-7}$                    | $\sim 0.1 - 1$ keV |
| B-class    | $10^{-7} - 10^{-6}$            | $\sim 1 - 5$ keV   |
| C-class    | $10^{-6} - 10^{-5}$            | $\sim 5 - 10$ keV  |
| M-class    | $10^{-5} - 10^{-4}$            | $\sim 10 - 25$ keV |

| Flare Type | X-ray Flux (W/m <sup>2</sup> ) | Energy Range (keV)   |
|------------|--------------------------------|----------------------|
| X-class    | $> 10^{-4}$                    | $\sim 25 - 100+$ keV |

Table 5.1 solar flare classification

The intensity and energy distribution of solar flare radiation directly influence the excitation of atomic shells and the resulting fluorescence emission.

## 5.2 Methodology

The data reduction pipeline was developed in Python, making extensive use of NumPy, SciPy, Astropy, and Matplotlib. The pipeline proceeds through following sequential stages, each of which is described in detail below, along with the scientific rationale.

### 5.2.1 Spectral Stacking

The fourteen spectrogram FITS files span the full month of April 2022 with each file covering a contiguous time interval. To enable uniform processing, all files were stacked sequentially along the time axis to produce a single month-long spectrogram array. The time array file was correspondingly concatenated.

$$S_{\text{stacked}}(t,C) = \text{concat}_t(S_1(t,C), S_2(t,C), \dots, S_{14}(t,C))$$

The result is saved as a NumPy binary file (stacked\_spectrogram.npy) for efficient random-access loading in subsequent steps.

### 5.2.2 Flare Event Identification

Using the enhancement file, the UTC timestamps of each recorded flare event were extracted. These intervals were matched against the stacked time array to identify which time indices fall within each flare window. This cross-matching was performed using a Boolean mask:

$$\text{event\_mask}(t) = \begin{cases} \text{True,} & \text{if } t_{\text{start\_flare}} \leq t_{\text{array}}(t) \leq t_{\text{end\_flare}} \\ \text{False,} & \text{otherwise} \end{cases}$$

The resulting event indices array contains the integer indices into the stacked spectrogram corresponding to flare-active observations. Only these indices are passed forward for spectral fitting. This step is the critical data-selection gate of the entire pipeline.

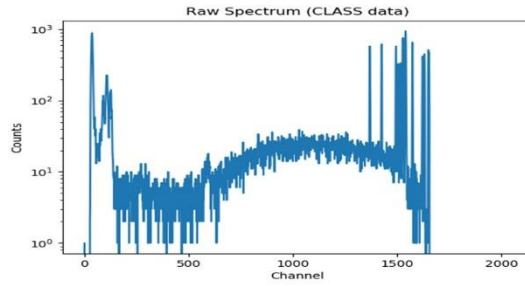


Fig 5.1 Raw spectrum of CLASS data

## 5.3 Energy Calibration

The CLASS instrument carries an on-board Fe-55 radioactive calibration source that produces characteristic X-ray lines at known energies — most importantly the Mn K-alpha line at 5.895 keV and the Mn K-beta line at 6.490 keV (Fe-55 decays by electron capture to Mn-55, which subsequently de-excites by emitting characteristic Mn X-rays). These lines appear as sharp, isolated Gaussian peaks in the detector spectrum and serve as fixed-energy reference points for calibration.

Prior to the analysis of any flare-selected spectra, the Fe-55 calibration spectra acquired at varying detector temperatures were individually fitted with Gaussian profiles. The resulting centroid channel positions of the Mn K-alpha and K-beta lines were recorded as a function of detector temperature. A linear regression through these calibration points establishes the channel-to-energy conversion factor (in eV per channel) and the zero-channel offset for each temperature regime. The calibration was found to converge to a value of 13.5 eV/channel as the adopted energy calibration representing the thermally-averaged gain of the detector over the observation period.

This pre-flight calibration approach using the on-board source is standard practice for space-borne X-ray spectrometers. It provides an absolute energy reference that is independent of the source spectrum being observed and compensates for the progressive gain drift that is characteristic of cooled solid-state detectors operated in the radiation environment of deep space.

Without Energy calibration spectral features are identified only by channel number but the detector specific quantity that may drift with temperature and radiation damage. Physical energy units allow comparison with laboratory XRF databases, Solar spectral models, and data from other instruments.

Following energy calibration, the principal XRF emission lines targeted in this study were assigned the following nominal channel positions and energy windows, as summarised in Table 4.2: after calibration Gaussian fitting were done the parameters are stored in files and used for the further calculations

| Element | K-alpha Line (keV) | Nominal Channel | Energy Range Used (keV) |
|---------|--------------------|-----------------|-------------------------|
| Mg      | 1.254              | ~93             | 1.15 – 1.35             |
| Al      | 1.487              | ~110            | 1.35 – 1.60             |
| Si      | 1.740              | ~129            | 1.60 – 1.90             |

Table 5.2: Energy calibration results for the principal XRF emission lines

These energy windows define the spectral regions extracted for Gaussian fitting in the subsequent step.

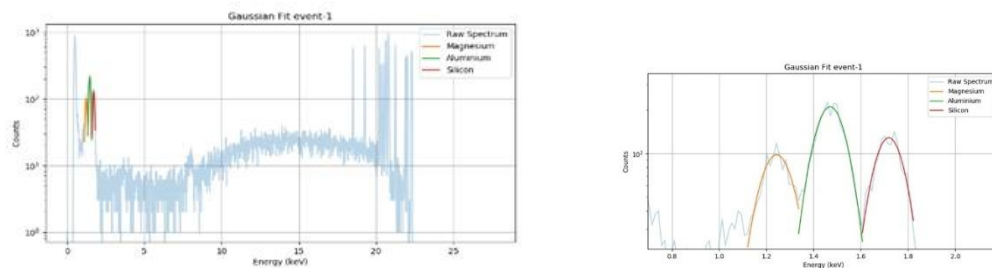


Fig 5.2 Energy fitted Gaussian curve the left fig shows the fit of Mg, Al, Si in enlarged view

## 5.5 Good vs. Bad Event Classification

Not all Gaussian fits converge to physically meaningful solutions. Fits can fail due to insufficient counts in a given accumulation window (low signal-to-noise), pile-up or saturation during intense flares, spectral confusion from overlapping lines, or numerical divergence of the fitting algorithm. A classification scheme was developed to distinguish good events (reliable fits) from bad events (unreliable or non-physical fits) based on the following criteria:

| Criterion          | Good Event Condition                     | Physical Rationale             |
|--------------------|------------------------------------------|--------------------------------|
| Line centroid (mu) | Within $\pm 0.05$ keV of tabulated value | Energy calibration consistency |

| Criterion          | Good Event Condition                           | Physical Rationale                    |
|--------------------|------------------------------------------------|---------------------------------------|
| Line width (sigma) | $0.03 \text{ keV} < \sigma < 0.25 \text{ keV}$ | Consistent with instrument resolution |
| Amplitude (A)      | $A > 3 \times \text{RMS\_noise}$               | Signal above 3-sigma noise floor      |
| Fit covariance     | No diagonal covariance elements = inf          | Convergence confirmation              |

Table 5.3: Good/bad event classification criteria.

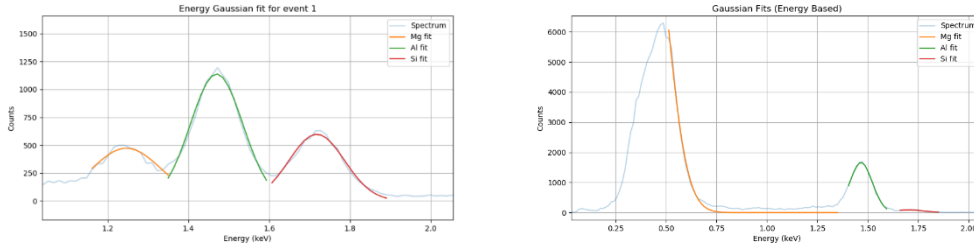


Fig 5.3 Good fit & Bad fit

Events failing any of the above criteria are flagged as bad events and excluded from the analysis. Good events form the final science dataset.

## 5.6 Adaptive Window Method

A critical challenge in CLASS data analysis is that different flare events have different durations and intensities. Fixing a rigid accumulation window leads to inconsistent signal-to-noise ratios: bright short flares may saturate, while faint long flares may not accumulate enough counts for a reliable fit within a 100-second window.

To address this, an adaptive window method was implemented. The algorithm operates as follows:

1. Initialise the window width  $W = W_{\text{min}} = 100$  seconds (one time bin).
2. Accumulate the spectrum over the current window  $[t_{\text{start}}, t_{\text{start}} + W]$ .
3. Attempt a Gaussian fit on the accumulated spectrum.
4. If the fit satisfies all good-event criteria: record fit parameters; advance start time to  $t_{\text{start}} + W$ ; reset  $W = W_{\text{min}}$ .
5. If the fit fails: extend the window by one bin ( $W += 100$  s); repeat from step 2.
6. Continue until the entire flare interval is processed.

The adaptive window method is equivalent to a variable-resolution temporal analysis: it automatically finds the shortest accumulation interval that yields a reliable spectral fit. This preserves temporal information where the signal is strong and avoids data loss where signal is weak.

**Window duration  $W(t)$ :  $W = n \times 100$  s,  $n = 1, 2, 3, \dots$**

*Note: The minimum window of 100 s is set by the CLASS native time resolution. In practice, most good fits occur within 1–4 bins (100–400 s) for moderate flares.*

after this method the line flux intensity were used to calculated as mention in chapter 4 (ref sec 4.11.1).

## 5.8 Geometry Merging

To introduce spatial context into the analysis, the good-event spectral results were merged with the SPICE geometry file. The merge was performed on the common time index, matching each good-event record to its corresponding latitude, longitude, and zenith angle.

The Solar zenith angle (SZA) must be converted to the solar incidence angle  $\theta_{inc}$

$$\theta_{inc} = 90^\circ - SZA$$

where the solar zenith angle (SZA) is defined as the angle between the local vertical (zenith) at a given point on a planetary surface and the direction to the Sun, and it is a key parameter controlling the intensity of incident solar radiation. When the Sun is directly overhead, the SZA is  $0^\circ$ , resulting in maximum irradiance, whereas at the horizon the angle approaches  $90^\circ$ , leading to minimal and highly oblique illumination.[22][23] After merging, each data record contains: timestamp, latitude, longitude, incidence angle, normalised line flux(cps) for each fitted line, fit quality flags. This merged dataset is the input for the spatial and polar analyses.

## 5.9 Analysis

A rectangular region of interest (ROI) in latitude-longitude space was defined based on the dense coverage visible in the spatial distribution plot (Chapter 5). Only good-event records falling within this ROI were used for the flux–angle correlation analysis, ensuring spatial homogeneity (i.e., the elemental abundance is approximately constant within the ROI, so flux variations are driven primarily by geometry).

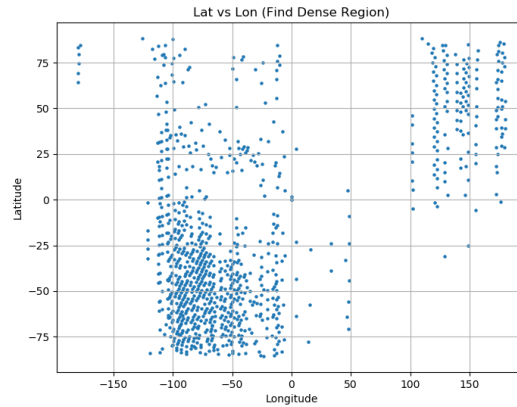


Fig 5.4 Rectangular ROI has good flux values

With in the selected ROI the incident beam angle vs Line flux intensity (cps) were plotted and The result line flux vs incident Angle plots are as below

### Magnesium

The following figure illustrates the plot between incident angle and Line flux (cps) of the Magnesium element.

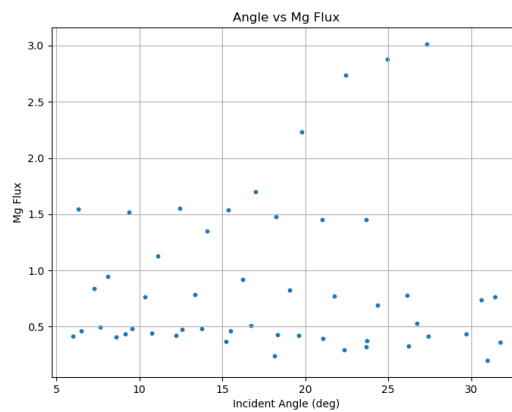


Figure 5.5 Incident angle vs. normalised Mg line flux (cps).

### Aluminium

The following figure indicates that the plot between incident angle and Line flux (cps) of the aluminium element. The figure shows that as the angle increases the line flux also increases

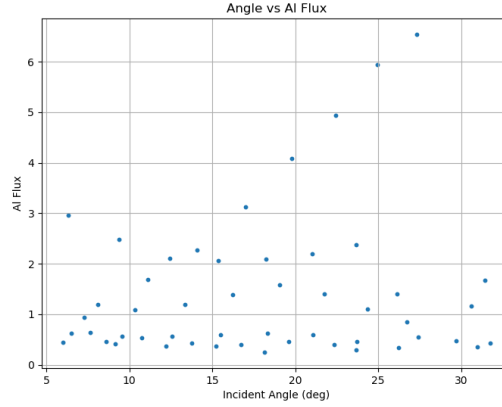


Figure 5.6 Incident angle vs. normalised Al line flux (cps).

## Silicon

The following figure illustrates the plot between incident angle and Line flux (cps) of the Silicon element.

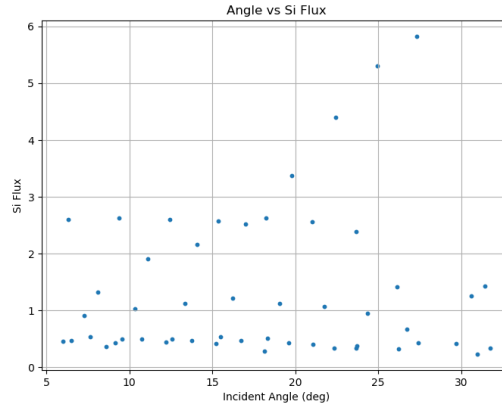


Figure 5.7 Incident angle vs. normalised Si line flux (cps).

## 5.10 Data Integration and Preprocessing (in wt%)

The analysis begins with two primary datasets obtained from the CLASS (Chandrayaan-2 Large Area Soft X-ray Spectrometer) payload. The first dataset contains spectral information for each observational event, including four latitude–longitude coordinate pairs defining the spatial footprint of the observation, along with normalized elemental flux values. The second dataset consists of derived elemental abundances expressed as weight percentages (wt%) along with their associated percentage uncertainties.

These two datasets are merged using a common event identifier, resulting in a unified dataset that contains, for each observation, spatial coordinates, spectral flux information, elemental

composition, SZA and corresponding measurement uncertainties. This merged dataset forms the foundation for subsequent geochemical analysis.

## 5.11 Reference Composition

The first step in the classification pipeline establishes a reference geochemical composition — a single representative point in multi-element chemical space against which all individual observations are compared. The reference is defined as the arithmetic mean of each elemental weight percentage across the entire dataset:

$$\bar{X}_{\text{ref}} = (1/N) \sum X_i$$

where  $\bar{X}_{\text{ref}}$  is the mean weight percentage of element X (computed separately for Mg, Al, Si, and Fe), N is the total number of spectral observations, and  $X_i$  is the elemental abundance of the  $i^{\text{th}}$  observation. Physically, this mean composition represents the average geochemical fingerprint of the observed lunar region, and serves as the centroid of the reference cluster in compositional space.

## 5.12 Distance in Multi-Element Compositional Space

Each individual spectral observation defines a point in four-dimensional geochemical space, with axes corresponding to the weight percentages of Mg, Al, and Si. The degree of geochemical similarity between an observation and the reference composition is quantified using the squared Euclidean distance:

$$d^2 = (\text{Mg} - \text{Mg}_{\text{ref}})^2 + (\text{Al} - \text{Al}_{\text{ref}})^2 + (\text{Si} - \text{Si}_{\text{ref}})^2$$

where  $d^2$  is the squared Euclidean distance of the observation from the reference point, and the subscripts denote the respective elemental weight percentages.

The Euclidean distance metric is physically appropriate in this context because each elemental axis carries comparable dimensional units (wt%). A small squared distance indicates compositional similarity to the average terrain a large squared distance flags an observation as geochemically anomalous or belonging to a distinct type. So the value of threshold is choosen as  $\alpha = 5 \text{ wt\%}$

## 5.14 Classification Criterion

Applying the threshold to the computed distance array yields a binary terrain classification for each observation:

$$\text{Class} = \text{Similar} \quad \text{if } d^2 < \alpha^2$$

$$\text{Class} = \text{Different} \quad \text{if } d^2 \geq \alpha^2$$

Geophysically, the 'Similar' class corresponds to observations consistent with the similar compositional area in the moon, while the 'Different' class captures observations characteristic of the contrasting areas in the moon

This classification framework effectively separates regions that are statistically indistinguishable from the reference composition from those that exhibit significant compositional anomalies.

## 5.15 Shoelace formula

The Shoelace formula (also known as the Surveyor's formula or Gauss's area formula) computes the signed area of a simple (non-self-intersecting) polygon with vertices listed in order — either clockwise or counter-clockwise — in a 2D Cartesian plane. For a quadrilateral with vertices  $(x_0, y_0)$ ,  $(x_1, y_1)$ ,  $(x_2, y_2)$ ,  $(x_3, y_3)$ , the formula takes the form:

$$A = \frac{1}{2} |(x_0y_1 - y_0x_1) + (x_1y_2 - y_1x_2) + (x_2y_3 - y_2x_3) + (x_3y_0 - y_3x_0)|$$

which can be written compactly in summation form as:

$$A = \frac{1}{2} | \sum_{k=0}^{n-1} (x_k y_{k+1} - x_{k+1} y_k) |$$

where the index  $k$  is taken modulo  $n$  (with  $n = 4$  for a quadrilateral), ensuring closure of the polygon. The absolute value ensures that the result is positive regardless of the vertex ordering convention (clockwise vs. counter-clockwise). The computed area  $A$  is in  $\text{km}^2$ , consistent with the projected coordinate units.

## 5.10 Incidence Angle vs Line Flux

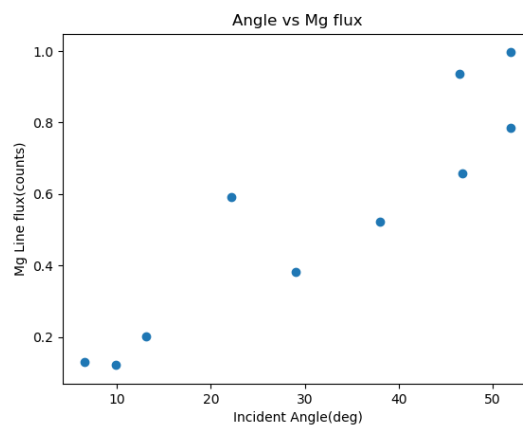
Physical interpretation: The positive trend is the fundamental observational confirmation of the XRF flux equation. Physically, when the Sun is nearly overhead (small  $\theta_{\text{inc}}$ ), the solar beam illuminates the surface at near-normal incidence. The projected beam area is small but the irradiance per unit area is high (approximately  $I_{\text{solar}}$ ). However,  $\cos(\theta_{\text{inc}})$  is small (close to zero for very small  $\theta_{\text{inc}}$ ), so the XRF production efficiency is low. As  $\theta_{\text{inc}}$  increases toward 50-60 degrees,  $\cos(\theta_{\text{inc}})$  approaches values that maximise the product  $I_{\text{solar}} \times \cos(\theta_{\text{inc}})$  relative to the noise floor, producing a stronger XRF signal. This is the same geometric effect as why surfaces facing the Sun more obliquely receive less heating per unit area — but in the XRF context it is the illumination of the subsurface that drives fluorescence.

The same positive trend was confirmed for the Si K-alpha and Mg K-alpha lines. This is expected from theory since  $\cos(\theta_{\text{inc}})$  enters the XRF equation multiplicatively and independently

of the element being observed. The fact that all three lines show the same trend provides a strong consistency check.

## Magnesium

The following figure shows the scatter plot of solar incidence angle  $\theta_{\text{inc}}$  (degrees) versus the normalised Al line flux (cps) for all good events in the polar region that passed the similarity filter. A clear positive correlation is observed: as  $\theta_{\text{inc}}$  increases from approximately  $5^\circ$  to  $53^\circ$ , the Al line flux also increases. It confirms that as the incident angle increases the line flux also increases there is a linear relation between them.



*Figure 5.1 Incident angle vs. normalised Mg line flux (cps).*

The positive correlation directly confirms the  $\cos \theta_{\text{inc}}$  geometric factor in the XRF equation. Each data point is one good-quality CLASS spectrum from within the polar region of interest.

## Aluminium

The following figure shows the scatter plot of solar incidence angle  $\theta_{\text{inc}}$  (degrees) versus the normalised Al line flux (cps) for all good events in the polar region that passed the similarity filter. A clear positive correlation is observed: as  $\theta_{\text{inc}}$  increases from approximately  $5^\circ$  to  $53^\circ$ , the Al line flux also increases from approximately 0.13 counts to approximately 1.35 counts. It confirms that as the incident angle increases the line flux also increases there is a linear relation between them.

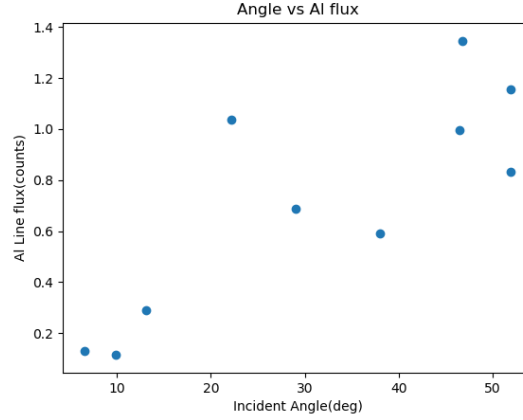
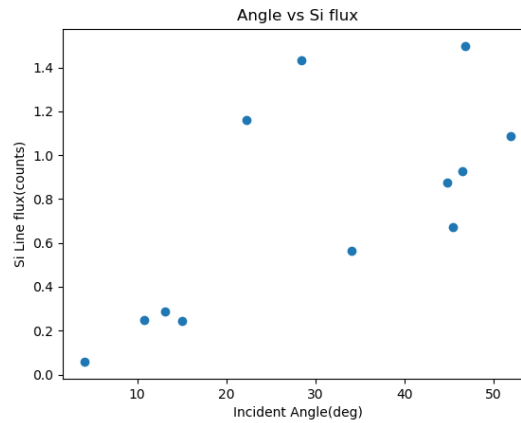


Figure 5.1 Incident angle vs. normalised Al line flux (cps) for CLASS polar region observations.

## Silicon

The following figure shows the scatter plot of solar incidence angle  $\theta_{\text{inc}}$  (degrees) versus the normalised Al line flux (cps) for all good events in the polar region that passed the similarity filter (as mentioned in chapter 4). The graph confirms that as the incident angle increases the line flux also increases there is a linear relation between them



The scatter trend reflects:

- (1) genuine compositional heterogeneity within the polar region — the analysis area is not perfectly homogeneous;
- (2) The scattering of abundance points is due different types of flares such as class C, class M flare etc
- (3) photon counting noise in individual CLASS spectra. After correcting for solar flux variation (future work), the residual scatter should reflect pure abundance variation

## 5.11 Spatial Distribution of Mg & Al Abundance

### Mg wt% vs. Al wt% — Mare-Highland Separation

The two-dimensional scatter plot of Magnesium weight percentage (Mg wt%) versus Aluminium weight percentage (Al wt%), derived from the CLASS dataset, constitutes a fundamental diagnostic tool for the geochemical characterisation of the lunar surface. Figure 5.1 presents this scatter plot for the subset of spectral observations classified as geochemically 'Similar' following the distance-based classification described in Section 5.8. The plot reveals two statistically and physically distinct populations in the Mg–Al compositional plane, corresponding to the two major terrain types on the lunar surface: the Lunar Highlands (Terraes) and the Lunar Maria.

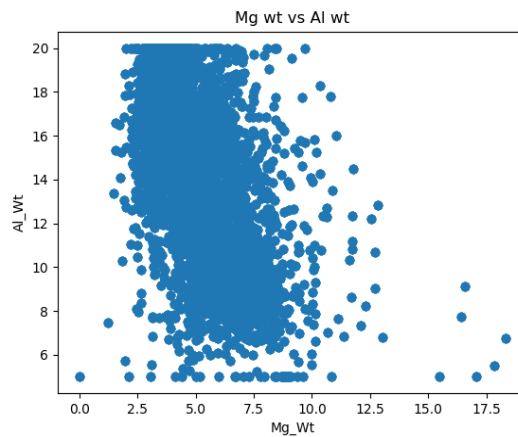


Figure 5.3: Scatter plot of Mg weight fraction (wt%) vs. Al weight fraction (wt%)

#### 5.10.1 Description of Observed Populations

##### *Population I — Dense Vertical Cluster: Lunar Highlands Signature*

The dominant feature of Figure 5.1 is a large, densely populated cluster of observations occupying the region Mg wt%  $\approx$  2–10 and Al wt%  $\approx$  5–20. This cluster exhibits a characteristic vertical elongation: the Aluminium abundance spans a wide range (5 to 20 wt%), while the Magnesium abundance remains constrained to consistently low values (2–10 wt%). The high point density within this cluster indicates that the majority of orbital observations in the dataset correspond to this compositional regime. This population is identified as the **Lunar Highland signature**.

##### *Population II — Sparse Rightward Scatter: Lunar Mare Signature*

A second, sparser population of observations is distributed toward higher Mg wt% values (Mg wt%  $\approx$  10–19) with moderate-to-low Al wt% (approximately 5–15 wt%). These points exhibit a weak negative diagonal trend in the Mg–Al plane — as Mg wt% increases, Al wt% tends to decrease —

and their sparse distribution reflects the comparatively limited areal coverage of mare terrain within the sampled orbital arc. This population corresponds to the **Lunar Mare signature**.

For example the latitude and longitude from the graph were picked and referred with the known values and it is confirmed that

Latitude: 72.3152, Longitude: 41.559 of the Moon - very close to North polar region represents the Aluminium rich highland were as Latitude – 31.1385, Longitude: -31.1743 falls on the near side of the moon is rich in Magnesium and it represents Mare region of the Moon.

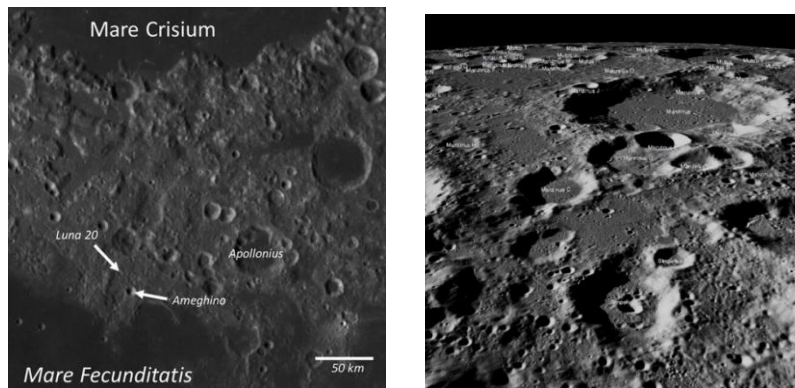


Fig 5. Shows the lunar Mare region and lunar Highland region

## 6 Conclusion

The conclusion of this thesis is that XRF line flux exhibits a strong, reproducible, and physically well-understood positive dependence on the solar incidence angle in both the controlled laboratory regime and the complex orbital environment of the Moon. This result, demonstrated across 15 distinct emission lines from 10 elements spanning atomic numbers  $Z = 12$  (Mg) to  $Z = 34$  (Se), establishes that geometric correction of orbital XRF data is not merely a calibration artefact but a physically necessary step rooted in the interaction-volume and path-length physics of the XRF process. Any quantitative abundance retrieval from CLASS data that neglects this angular modulation will introduce systematic errors of the order of 50–300% depending on the incidence angle range sampled — errors that are entirely avoidable once the empirical angular-flux relationship established in this thesis is applied.

The second major conclusion is that the CLASS instrument, operated in conjunction with the solar flare enhancement catalogue and the adaptive accumulation methodology developed here, yields physically meaningful, high-confidence XRF spectral fits for Mg  $K\alpha$ , Al  $K\alpha$ , and Si  $K\alpha$  — the three lightest elements detectable by CLASS — even under moderate C- and M-class solar flare conditions. The good/bad event classification framework, with its physically motivated centroid, sigma, amplitude, and covariance convergence criteria, provides a rigorous statistical basis for separating reliable fits from noise-dominated spectra, ensuring that only robust measurements enter the calibration database. The adaptive window algorithm maximises temporal coverage by automatically tuning the accumulation window to the local signal-to-noise ratio, extracting more science events per flare than a fixed-window approach.

The third conclusion concerns geochemical discrimination: the CLASS April 2022 dataset, processed through the complete pipeline described in this thesis, successfully reproduces the known mare-highland compositional dichotomy of the lunar surface in the Mg–Al abundance plane, providing an independent validation of the CLASS calibration and the abundance retrieval methodology. This result demonstrates that CLASS flare-time XRF measurements carry quantitatively meaningful geochemical information at a spatial resolution of  $\sim 12.5$  km — sufficient to distinguish individual mare basalt units and highland massifs.

## **7. Future Scope and Research Directions**

### **7.1 Solar Flux Normalisation Using XSM**

The most important extension of the present work is the incorporation of simultaneous solar flux measurements from the X-ray Solar Monitor (XSM) to normalise the CLASS line fluxes by the incident photon flux. In the current analysis, the raw line flux (cps) was used as the observable without accounting for the variability of the solar X-ray irradiance between different flare events. Since C-class and M-class flares differ in peak flux by one to two orders of magnitude, XSM-based normalisation will dramatically reduce the scatter in the incidence-angle versus line-flux plots — scatter that was explicitly attributed in Chapter 5 to inter-flare irradiance variation rather than compositional heterogeneity. Once implemented, the normalised flux will be proportional solely to the elemental abundance and the incidence-angle geometric factor, enabling a clean, single-parameter empirical calibration.

### **7.2 Extension to Higher Solar Activity Periods**

The current analysis was restricted to April 2022, a period of moderate solar activity containing primarily C- and M-class flares. Extending the pipeline to X-class flare periods — particularly as the Sun approaches solar maximum (predicted circa 2025) — will provide CLASS spectra with substantially higher signal-to-noise ratios, enabling detection of weaker emission lines such as Ca K $\alpha$  (3.69 keV), Ti K $\alpha$  (4.51 keV), and Fe K $\alpha$  (6.40 keV) with sufficient counts for robust Gaussian fitting. The energy range above 3 keV is critical for distinguishing mafic (Fe-Ti-rich) from felsic (Ca-Al-rich) terrain types and for identifying cryptomare units concealed beneath thin highland ejecta blankets. A larger, higher-quality dataset will also enable statistically robust investigation of secondary XRF enhancement effects — the re-excitation of one element's fluorescence by the characteristic X-rays of another — which are non-negligible in the multi-element lunar regolith matrix.

### **7.3 Full Elemental Abundance Map Generation**

The ultimate scientific goal of CLASS is to produce pixel-by-pixel elemental abundance maps of the entire lunar surface at 12.5 km resolution. The empirical calibration relations established in this thesis — linking measured line flux to elemental weight percent as a function of incidence angle — provide the mathematical foundation for this mapping exercise. Future work should integrate the geometry-corrected, XSM-normalised line fluxes with a full forward model of the XRF production process (incorporating the Sherman equation,

Beer-Lambert attenuation, matrix correction factors, and the CLASS instrument response matrix) to invert the measured flux into absolute abundance maps. Such maps will complement existing datasets from Kaguya's XRS, MESSENGER's XRS, and Apollo sample-return geochemistry, providing a uniquely high-resolution view of the Moon's surface composition that can constrain models of the lunar magma ocean, the chronology of mare volcanism, and the distribution of KREEP-rich terrains in the Procellarum KREEP Terrane.

## **7.4 Future Planetary Missions**

The dual-phase laboratory-orbital methodology developed in this thesis is directly transferable to future XRF missions targeting other airless bodies. Mercury, whose surface elemental composition has been partially mapped by MESSENGER's XRS, remains a priority target for high-resolution geochemical mapping by the BepiColombo MIXS (Mercury Imaging X-ray Spectrometer) instrument. The empirical angular-flux calibration framework and the adaptive spectral fitting pipeline developed here can be adapted to MIXS data with minimal modification, since the fundamental physics of solar-induced XRF is identical across airless bodies — only the solar irradiance spectrum, the target composition, and the orbital geometry differ. Similarly, NASA's planned Lunar-VISE and CLPS missions, as well as proposed asteroid sample-return XRF experiments, will benefit from the geometric correction methodology and good/bad event classification criteria established here. Beyond planetary science, the laboratory angular-XRF methodology has immediate applications in materials characterisation, thin-film analysis, and non-destructive testing scenarios where grazing-incidence XRF (GIXRF) is used to probe sub-micrometre surface layers.

## 8 Reference

- [1] Arne Hessenbruch, “A brief history of x-rays”.
- [2] Beckhoff, B., et al. (2006). *Handbook of Practical X-Ray Fluorescence Analysis*. Springer.
- [3] Peter. Brouwer, Theory of XEF: getting acquainted with the principles. PANalytical BV, 2003.
- [4] C. Suryanarayana and M. G. Norton, “X-Rays and Diffraction,” in X-Ray Diffraction, Boston, MA: Springer US, 1998, pp. 3–19. doi: 10.1007/978-1-4899-0148-4\_1.
- [5] Van Grieken, R., & Markowicz, A. (2002). *Handbook of X-ray Spectrometry*. Marcel Dekker.
- [6] Jenkins, R., Gould, R. W., & Gedcke, D. (1995). *Quantitative X-ray Spectrometry*. Marcel Dekker
- [7] Leonid V. Azároff, “Theory of Extended Fine Structure of X-Ray Absorption Edges”.
- [8] K. Ogawa, T. Okada, K. Shirai, and M. Kato, “Numerical estimation of lunar X-ray emission for X-ray spectrometer onboard SELENE,” 2008
- [4] A. Assmus, “Early History of X Rays.”
- [5] W. H. Bragg, W. L. Bragg Apr, B. W. H Bragg, and C. Professor of Physics, “The Reflection of X-rays by Crystals.” [Online]. Available: <https://royalsocietypublishing.org/>
- [7] Leonid V. Azároff, “Theory of Extended Fine Structure of X-Ray Absorption Edges”.
- [8] K. Ogawa, T. Okada, K. Shirai, and M. Kato, “Numerical estimation of lunar X-ray emission for X-ray spectrometer onboard SELENE,” 2008.
- [9] I. Waller and Douglas Rayner Hartree, “On the intensity of total scattering of Xrays”.
- [10] M. A. Blokhin, “TRAN SERI THE PHYSICS OF X-RAYS Second, Revised Edition A Office of Technical Information.”
- [11] Friso van der Veen and Franz Pfeiffer, “Coherent x-ray scattering”.
- [12] Keith A. Nugent, “Coherent methods in the X-ray sciences”.
- [13] R. A. Bonham, “Corrections to the Incoherent Scattering Factors for Electrons and X Rays”.
- [14] P.P. Kane, “Inelastic scattering of X-rays and gamma rays”.

- [15] Peter Rez, “Coherent and incoherent imaging of biological specimens with electrons and X-rays”.
- [16] S. K. Sinha M. Tolan A. Gibaud, “Effects of partial coherence on the scattering of x rays by matter”.
- [17] D. V. Rao, G. E. Gigante, and R. Cesareo, “Coherent and incoherent scattering of 42.75 and 47.24 keV x-ray photons scattered from Al, Cu, Y, Mo, Au and Pb,” XRay Spectrometry, vol. 24, no. 4, pp. 172–176, Jul. 1995, doi: 10.1002/xrs.1300240406.
- [18] N. Broll, “Quantitative x-ray fluorescence analysis. Theory and practice of the fundamental coefficient method”.
- [19] Handbook of Chandrayaan 2 Payloads Data & Science - Indian Space Research Organisation Bengaluru
- [20] Knoll, G. F., *Radiation Detection and Measurement*, 4th Edition, Wiley, 2010.
- [21] Gatti, E., and Rehak, P., “Semiconductor Drift Chamber,” *Nucl. Instrum. Methods Phys. Res. A*, 225, 1984.
- [22] Liou, K. N. (2002). *An Introduction to Atmospheric Radiation* (2nd ed.). Academic Press.
- [23] Hapke, B. (2012). *Theory of Reflectance and Emittance Spectroscopy* (2nd ed.). Cambridge University Press.